

Partial Results on the WOWII-133 Graffiti Conjecture

The C_4 -free branch below $\lfloor \ell \rfloor = 4$

Haoyu Chen*

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Abstract

Conjecture 133 of the *Written on the Wall II* series, produced by the Graffiti.pc conjecture-making program, asserts that a connected graph G satisfies $\text{rad}(G) + \lfloor \ell(G) \rfloor^c \leq \text{path}(G)$, where $\text{path}(G)$ is the order of a longest induced path, $\ell(G)$ is the average over v of the independence number of the open neighbourhood $N(v)$, and $c = 1$ if G is C_4 -free and $c = 0$ otherwise. We report partial progress. The C_4 -containing branch and the whole range $\lfloor \ell \rfloor \leq 2$ are settled unconditionally. For the first open band $\lfloor \ell \rfloor = 3$ we prove that a counterexample of minimum order must have radius at least 4: the radius-2 layer is closed by a fibre-counting theorem, and the radius-3 layer is closed by a rigidity analysis of the induced 6-cycle that such a counterexample is forced to carry, whose engine is an unconditional rigidity lemma — in a C_4 -free graph with no induced P_6 , a vertex outside an induced C_6 has 0 or 2 neighbours on the cycle, and if 2 they are antipodal. We also prove the $D = 2$ case of the natural strengthening $\text{path} \geq \text{diam} + 4$ for C_4 -free graphs of diameter D with $\ell > 3$, and, in the band $\lfloor \ell \rfloor \geq 4$, that a C_4 -free graph all of whose peripheral geodesics have low-independence side-neighbours satisfies $\ell < 4$. A deliberate part of the report is negative: we record an infinite family showing that no two-endpoint local extension lemma can reach $\text{rad} + k$ for $k \geq 4$, a 150-vertex witness killing the layered-counting route at radius 3, and two counterexamples showing that the deep endpoint-extension lemma available at geodesic length $d \geq 4$ is false at $d = 3$ both with and without the natural repair hypothesis. Every statement carries an explicit epistemic marker, and the verification scripts are inventoried in an appendix.

***Disclosure of the production pipeline.** This paper reports the certified by-products of an autonomous, still-running attack on Conjecture 133 of the *Written on the Wall II* list. *The conjecture is not resolved here, and this paper does not claim to resolve it.* Every mathematical statement below was produced by automated systems: an orchestrating agent (Claude Fable 5) that set the attack targets, adjudicated the returning reports line by line, executed the promotion protocol of §1.3 and drafted this document; a free exploration channel (Qwen3.8-Max, web interface) that was given self-contained task briefs and whose output was never adopted without independent re-derivation; adversarial reviewers drawn from a model family disjoint from the one that produced the argument under review (Claude Opus-family instances and Qwen3.8-Max, each run blind or diff-scoped, in isolation, forbidden to consult the project’s other files); and machine checkers (purpose-written C and Python enumerators and independent from-scratch re-verifiers). No human supplied a definition, a lemma, a proof idea, or a correction. Provenance is recorded per result in §1.4: several of the arguments below originated in the free channel and were re-proved before being admitted, and one of the paper’s results — Theorem 9.4, a refutation — was found by a reviewer whose assignment was to look for defects. Several of the campaign’s own working claims were refuted by its own later rounds; those self-refutations are recorded in §9 rather than quietly deleted. Responsibility for what follows rests with the reader who re-runs the scripts of Appendix A.

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1 Introduction

1.1 The conjecture

Throughout, G is a finite simple graph on $n = |V(G)| \geq 2$ vertices with m edges and T triangles, $d(v)$ is the degree of v , $t(v)$ the number of triangles through v , δ and Δ the minimum and maximum degree, rad , diam , ecc the radius, diameter and eccentricity, and $N(v)$ the *open* neighbourhood of v . We write

$$a(v) := \alpha(G[N(v)]), \quad \ell(G) := \frac{1}{n} \sum_{v \in V} a(v), \quad \mu(G) := \min_v a(v),$$

where α is the independence number, and $\text{path}(G)$ for the number of *vertices* of a longest induced (chordless) path of G . We call $a(v)$ the *a-value* of v and ℓ the *neighbourhood independence average*.

We take the statement of the conjecture from its Lean transcription in the `formal-conjectures` corpus, file `FormalConjectures/ WrittenOnTheWallIII/ GraphConjecture133.lean`, where it is recorded as conjecture 133 of the *Written on the Wall II* series produced by the Graffiti.pc conjecture-making program of E. DeLaViña. We have not consulted the original *Written on the Wall II* document; the numbering and the attribution here are those of the Lean file, and the reader who needs the primary source should check it. The Lean statement reads

```
theorem conjecture133 (G : SimpleGraph α) [DecidableRel G.Adj]
  (h : G.Connected) :
  let rad := G.radius.toNat
  let hasC4 := exists a b c d, (pairwise distinct) and
    G.Adj a b and G.Adj b c and G.Adj c d and G.Adj d a
  let cC4 : ℕ := if hasC4 then 0 else 1
  (rad : ℝ) + (floor (1 / G) : ℝ) ^ cC4 <= (path G : ℝ)
```

Three points of the transcription matter and were checked against the Lean sources before any mathematics was attempted. First, `path` is the induced-path order, not the “average distance” homonym that the same corpus uses elsewhere. Second, `hasC4` asks for a 4-cycle as a *subgraph*, chords allowed; equivalently, G fails to be C_4 -free exactly when some two distinct vertices have two common neighbours. This *strong* form of C_4 -freeness is the one used everywhere below, and the distinction is load-bearing: several lemmas are false under the induced-only reading. Third, the exponent $c \in \{0, 1\}$ is applied through `Monoid.npow`, so on the C_4 -containing branch the right-hand side is $\text{rad} + 1$ whatever ℓ is.

Both sides being integers cast into $\mathbb{R}$, the Lean goal is equivalent to the conjunction of

- **(C133-a)** G connected with a 4-cycle: $\text{path}(G) \geq \text{rad}(G) + 1$;
- **(C133-b)** G connected and C_4 -free: $\text{path}(G) \geq \text{rad}(G) + \lfloor \ell(G) \rfloor$.

1.2 What is proved here, and what is not

The conjecture is open. Table 1 is the honest ledger; the numbers in it are justified in the sections named.

The main contributions are these.

1. **The closed half, with sharp constants.** Theorem 3.1 settles (C133-a) in four lines. Theorem 3.4 — every connected C_4 -free graph of radius ≥ 2 has $\text{path} \geq \text{rad} + 2$ — settles $\lfloor \ell \rfloor \leq 2$. Both are tight on the five extremal graphs K_2, K_3, C_5, C_6 and Petersen (§3.1), which is what forbids any argument that loses a constant.
2. **An endpoint-extension calculus and the band $\lfloor \ell \rfloor = 3$ with no $a = 1$ vertex.** Lemmas 4.1, 4.2, 4.3 extend a geodesic by one or two vertices from data at its ends. Theorem 4.5 combines them: $\mu \geq 2$ together with one vertex of a -value ≥ 3 forces $\text{path} \geq \text{rad} + 3$. This alone closes every C_4 -free graph on at most 13 vertices (§3.1).

Region	Status	Where
G contains a 4-cycle	closed	Thm. 3.1
C_4 -free, $\lfloor \ell \rfloor \leq 1$	closed	Thm. 3.2
C_4 -free, $\lfloor \ell \rfloor = 2$	closed	Thm. 3.4, Cor. 3.5
C_4 -free, $\lfloor \ell \rfloor = 3, \mu \geq 2$	closed	Thm. 4.5, Cor. 4.6
C_4 -free, $\lfloor \ell \rfloor = 3, \text{rad} = 2$	closed	Thm. 5.3, Cor. 5.5
C_4 -free, $\lfloor \ell \rfloor = 3, \text{rad} = 3$	closed <i>for a minimum-order counterexample</i>	Thm. 7.1
C_4 -free, $\lfloor \ell \rfloor = 3, \text{rad} \geq 4$	open ; reduced to Problem 6.6	§6
C_4 -free, $\lfloor \ell \rfloor \geq 4$	open ; partial results in §8	§8

Table 1: Status of Conjecture 133 after the work reported here.

3. **The radius-2 layer.** Theorem 5.3: a connected C_4 -free graph with $\text{rad} = 2$ and $\ell \geq 3$ always contains a pair of vertices at distance 2 with a -values $(\geq 3, \geq 2)$. The proof is a fibre decomposition of the ball of radius 2 around a high vertex followed by an exact count. Together with Lemma 4.2 this closes $\text{rad} = 2$.
4. **The radius-3 layer** (§7, the longest part of the paper). A minimum-order counterexample of radius 3 is reduced — by a peeling induction whose residual is made explicit — to a graph C that is C_4 -free, has $\text{rad}(C) = 2$, $\text{diam}(C) = 3$, $\ell(C) > 3$, no induced P_6 , and the extra property (R3b) that every vertex of a -value ≥ 3 has eccentricity ≤ 2 . Two theorems empty that class. Theorem 7.13 shows no vertex of such a graph has a -value 1 and eccentricity 3; Theorem 7.23 then shows the surviving branch is empty, by way of Lemma 7.18, an unconditional rigidity fact about induced 6-cycles in P_6 -free C_4 -free graphs which collapses the configuration to at most nine vertices.
5. **The integer form of the floor-free strengthening at $D = 2$.** Theorem 6.7: C_4 -free, $\text{diam} = 2$, $\ell > 3$ imply $\text{path} \geq 6 = \text{rad} + 4$. This is a non-trivial instance of the real-valued strengthening $\ell \leq \text{path} - \text{rad}$ (§3.2) and is the $D = 2$ case of Problem 6.6, to which all of $\text{rad} \geq 4$ has been reduced.
6. **The band $\lfloor \ell \rfloor \geq 4$** (§8). Three results at the lower reviewed tier: the deep endpoint-extension lemma known at geodesic length $d \geq 5$ is extended to $d = 4$ unconditionally (Lemma 8.1) and to $d = 3$ under one extra hypothesis (Lemma 8.2); and Theorem 8.4, a global counting theorem: a connected C_4 -free graph in which every geodesic of length $\geq \text{rad}$ has all its usable side-neighbours of a -value ≤ 3 satisfies $\ell < 4$. Contrapositively, every C_4 -free graph with $\ell > 4$ supplies, at every radius, a peripheral geodesic with a side-neighbour of a -value ≥ 4 .
7. **Negative results** (§9), reported because they close routes. An infinite family T_m of trees shows that “both endpoints of a -value $\geq k$ implies $\text{path} \geq \text{rad} + k$ ” is false for every $k \geq 4$. A 150-vertex graph (Hoffman–Singleton with two pendant vertices at each vertex) has $\lfloor \ell \rfloor = 3$, radius 3, and *no* pair at distance $\geq \text{rad}$ with a -values $(\geq 3, \geq 2)$, which kills the layered-counting route above radius 2. Two ten-vertex graphs refute the $d = 3$ analogue of Lemma 8.1, one without and one with the natural repair hypothesis (Theorem 9.4); the conclusion is that the free local route at $d = 3$ is closed and a global hypothesis must be paid for. Finally we record two defects that the review protocol found in the project’s own earlier text — a statement stated more widely than its proof supported, and an arithmetic error in the peeling bookkeeping — together with their repairs, since both are instructive.

Nothing here is formalised in Lean, and no claim below should be read as a formal verification. The scripts of Appendix A check assertions over stated finite ranges and on named explicit graphs; the proofs are ordinary mathematics.

1.3 Verification conventions and epistemic markers

Because this is an automated campaign’s output, the status of each statement is recorded explicitly and is *not* uniform. Four markers are used, attached to every numbered assertion; definitions and open problems assert nothing and carry none.

- [proved here; assertions machine-checked by `script`] — the proof is reproduced here in full, and the assertions it makes are checked by the named script over exactly the range stated at the point of use, usually an exhaustive table of small graphs plus explicit large witnesses. A script never checks more than the statement it is attached to.
- [proved here; machine-checked by `script`; **two-family review**] — as above, and in addition the statement passed the campaign’s *two-family* adversarial protocol: an independent reviewer from a model family disjoint from the one that produced the argument, run blind (given the text and nothing else, with no access to the project’s other files and no internet, permitted to write and run its own verification code), *plus* a second, diff-scoped pass over the same material by a different reviewer, both returning no surviving defect at any joint classified as mathematics. Every such statement in this paper survived at least one reviewer finding a *real* defect elsewhere in its neighbourhood; the repairs are in the text.
- [machine-assisted proof, re-derived here; machine-checked by `script`; **one adversarial review pass**] — *machine-assisted*: the argument originated in an automated solve attempt in the free exploration channel, was re-derived line by line here before admission (and in two places repaired — the repairs are marked in the proofs), is machine-checked as above, and passed *one* adversarial review pass returning no mathematics defect. This is a strictly weaker tier than [proved here; machine-checked by ; **two-family review**]: one review family, not two. The three results of §8 carry it, and the reader should treat them accordingly.
- [machine-verified: `script`] — a purely computational statement: an explicit finite graph together with its invariants, or a completed bounded sweep. It is evidence of exactly the stated scope and of nothing beyond it. In particular a *failed* bounded search is never a proof of impossibility, and §9 records the one occasion on which the campaign misreported such a search: a sweep over 1112 local *frames* that found no counterexample was written up in a way that read as evidence about *graphs*, and a graph counterexample was found later (Theorem 9.4).

Two further conventions. No result below was obtained with a SAT solver; the only exhaustive searches are over labelled small-graph tables (explicitly stated), over bounded local configurations with all edge patterns, and over the neighbourhoods of named explicit graphs. And where a numerical claim is a *lower bound* on path obtained by an early-exit search, we say “path $\geq k$ ” rather than “path = k ”.

1.4 Provenance of the arguments

The results below were produced by three channels, and it matters which.

The orchestrating agent produced, and is solely responsible for, Lemmas 4.1–4.3, Theorems 4.5, 6.7, Lemmas 7.16–7.22, Theorems 7.13 and 7.23, the reductions of §6, and every negative result of §9 except Theorem 9.4.

The free exploration channel produced first drafts of the extension lemma Lemma 4.2, of the radius-2 theorem Theorem 5.3, and of the three results of §8. None was adopted as written: each was re-derived independently before admission, three claimed solutions of the residual problem of §7 were checked and *rejected*, and the eventual proof of Theorem 7.23 runs by a route (Lemma 7.18) that the free channel never produced. Two garbled justifications inside the arguments of §8 were replaced by correct ones during re-derivation and are marked in place.

The adversarial reviewers produced no positive result by design, with one exception: the reviewer assigned to §8 constructed the graph H of Theorem 9.4, which refutes a target that the orchestrating agent’s own sweep had recorded as open. That is recorded here as a result of the review channel.

1.5 Related routes and what is known to be impossible

Section 9 lists five named obstacles, of which one deserves mention up front. Every bound proved in §3 is driven by δ or by rad , and both are destroyed by a single local modification while ℓ moves by $O(1/n)$: Hoffman–Singleton with one edge subdivided is C_4 -free with $n = 51$, $\delta = 2$, $\text{rad} = 3$, $\ell = 6.902$, so that $\max(\text{rad} + 2, \delta + 2, \text{rad} + \lfloor \delta/2 \rfloor) = 5$ while the conjecture demands 9. Hence Theorems 3.4, 3.6, 3.7 *cannot* be combined into a proof; a correct proof must use a quantity as robust as ℓ itself. That observation is why the campaign moved to the a -value calculus of §4.

2 Preliminaries

Lemma 2.1 (local matching). *[proved here; assertions machine-checked by `wowii133_verify.c`] Let G be C_4 -free. Then for every v the graph $G[N(v)]$ has maximum degree ≤ 1 , i.e. it is a disjoint union of $t(v)$ edges and $d(v) - 2t(v)$ isolated vertices. Consequently*

$$a(v) = d(v) - t(v), \quad \lfloor d(v)/2 \rfloor \leq a(v) \leq d(v), \quad \ell(G) = \frac{2m - 3T}{n}.$$

Proof. If $u \in N(v)$ had two neighbours $w_1, w_2 \in N(v)$ then vw_1uw_2v is a 4-cycle on four distinct vertices. So $G[N(v)]$ is a matching plus isolated vertices; an edge inside $N(v)$ is exactly a triangle through v , so there are $t(v)$ of them. A maximum independent set takes one endpoint of each edge and every isolated vertex, giving $t(v) + (d(v) - 2t(v)) = d(v) - t(v)$. Summing and using $\sum_v t(v) = 3T$, $\sum_v d(v) = 2m$ gives the last identity. $\square$

So (C133-b) is the statement $\text{path}(G) \geq \text{rad}(G) + \lfloor (2m - 3T)/n \rfloor$; for graphs of girth ≥ 5 it is $\text{path} \geq \text{rad} + \lfloor \bar{d} \rfloor$ with $\bar{d}$ the average degree. Note $\ell \geq 1$ always, so the open region is $\lfloor \ell \rfloor \geq 3$, i.e. $2m - 3T \geq 3n$.

The following vocabulary is used throughout. For v in a C_4 -free graph, the *components of $N(v)$* always means the components of $G[N(v)]$, which by Lemma 2.1 are single vertices and single edges; their number is exactly $a(v)$. A *geodesic* $u_0 \dots u_d$ is a shortest path, hence automatically induced. A vertex $x \in N(u_0)$ is a *usable side-neighbour* of the end u_0 of that geodesic if x lies outside u_1 ’s component of $N(u_0)$; one exists iff $a(u_0) \geq 2$, and a usable side-neighbour satisfies $x \not\sim u_1$ by definition. A vertex h with $a(h) \geq 3$ is called *high*, or a *hub*, and $H := \{v : a(v) \geq 3\}$.

Lemma 2.2 (classification of $a = 1$). *[proved here; assertions machine-checked by wowii133_gplus.c]* Let G be C_4 -free. Then $a(v) = 1$ if and only if $N(v)$ is a clique of size ≤ 2 , i.e. v is a leaf, or $d(v) = 2$ and the two neighbours of v are adjacent. We call the latter a triangle-leaf.

Proof. $a(v)$ is the number of components of $G[N(v)]$, which is a matching plus isolated vertices; it equals 1 iff there is exactly one component, i.e. one vertex or one edge. $\square$

3 The closed part of the conjecture

Theorem 3.1 (the C_4 -containing branch). *[proved here; assertions machine-checked by wowii133_verify.c]* For every connected G , $\text{path}(G) \geq \text{rad}(G) + 1$. In particular (C133-a) holds.

Proof. Let c be a centre, $\text{ecc}(c) = r = \text{rad}$, and let z satisfy $\text{dist}(c, z) = r$. A shortest c - z path $u_0 \dots u_r$ is induced, since a chord $u_i u_j$ with $j \geq i + 2$ would give a shorter c - z walk. It has $r + 1$ vertices. $\square$

Theorem 3.2. *[proved here]* If $\lfloor \ell(G) \rfloor \leq 1$ then (C133-b) holds.

Proof. Immediate from Theorem 3.1. $\square$

Theorem 3.3 (radius ≤ 1). *[proved here; assertions machine-checked by wowii133_verify.c]* If G is connected, C_4 -free, $n \geq 2$ and $\text{rad}(G) \leq 1$, then $\ell(G) < 2$; hence $\lfloor \ell(G) \rfloor = 1$ and (C133-b) holds.

Proof. Let v dominate, so $N(v) = V \setminus \{v\}$. By Lemma 2.1 $G - v$ is a matching: say a edges and b isolated vertices, $n = 1 + 2a + b$. Then $a(v) = a + b$; a vertex u matched to u' has $N(u) = \{v, u'\}$ with $v \sim u'$, so $a(u) = 1$; an unmatched u has $N(u) = \{v\}$, $a(u) = 1$. Hence $\sum_w a(w) = (a + b) + 2a + b = 3a + 2b < 2(1 + 2a + b) = 2n$. $\square$

Theorem 3.4 (radius ≥ 2). *[proved here; assertions machine-checked by wowii133_verify.c]* If G is connected, C_4 -free and $\text{rad}(G) \geq 2$, then $\text{path}(G) \geq \text{rad}(G) + 2$.

Proof. Write $r = \text{rad}(G)$.

Case 1: $\text{diam}(G) \geq r + 1$. A geodesic between two vertices at distance diam is induced and has $\text{diam} + 1 \geq r + 2$ vertices.

Case 2: $\text{diam}(G) = r$. Since $r \leq \text{ecc}(v) \leq \text{diam}$ for every v , every vertex is a centre.

Case 2a: some vertex c has $d(c) \geq 3$. Pick z with $\text{dist}(c, z) = \text{ecc}(c) = r \geq 2$ and a geodesic $c = u_0, u_1, \dots, u_r = z$. At most one vertex of $N(c) \setminus \{u_1\}$ is adjacent to u_1 : two such vertices x, x' would give the 4-cycle cxu_1x' . As $|N(c) \setminus \{u_1\}| \geq 2$, choose $x \in N(c) \setminus \{u_1\}$ with $x \not\sim u_1$. Then $x, c, u_1, \dots, u_r$ is an induced path on $r + 2$ vertices: $x \neq u_i$ for all i ; $x \not\sim u_1$ by choice; $x \not\sim u_2$, else cxu_2u_1 is a 4-cycle; and $x \not\sim u_i$ for $i \geq 3$ since $\text{dist}(x, u_i) \geq i - 1 \geq 2$.

Case 2b: $\Delta \leq 2$. Then G is a path or a cycle. A path P_n has $\text{diam} = n - 1$ and $\text{rad} = \lceil (n - 1)/2 \rceil$, so $\text{diam} = \text{rad}$ forces $n \leq 2$, contradicting $\text{rad} \geq 2$. So $G = C_n$; C_3 has radius 1 and C_4 is not C_4 -free, so $n \geq 5$, and $\text{path}(C_n) = n - 1 \geq \lfloor n/2 \rfloor + 2 = \text{rad} + 2$. $\square$

Corollary 3.5. *[proved here]* Conjecture 133 holds for every connected graph containing a 4-cycle, and for every connected C_4 -free graph with $\lfloor \ell(G) \rfloor \leq 2$.

Proof. Theorem 3.1; and for C_4 -free G : if $\lfloor \ell \rfloor \leq 1$ use Theorem 3.2; if $\lfloor \ell \rfloor = 2$ then $\text{rad} \geq 2$ by Theorem 3.3, and Theorem 3.4 applies. $\square$

Theorem 3.4 is tight on C_5 and C_6 . Two greedy bounds complete the picture of what elementary arguments give.

Theorem 3.6 (greedy induced path). *[proved here; assertions machine-checked by wowii133_verify.c]* (E1) *If G is C_4 -free and triangle-free (girth ≥ 5) with $\delta \geq 2$, then $\text{path}(G) \geq \delta + 2$.* (E2) *If G is C_4 -free with $\delta \geq 3$, then $\text{path}(G) \geq \delta + 1$.*

Proof. Grow an induced path $v_1 \dots v_k$ greedily. A vertex $x \in N(v_k)$ fails to extend it iff x lies on the path or $x \sim v_i$ for some $i \leq k - 1$. Count failures: v_{k-1} is the only path vertex in $N(v_k)$; a common neighbour of v_{k-1} and v_k is a triangle-mate, of which C_4 -freeness allows at most one, and none if G is triangle-free; v_{k-1} is already a common neighbour of v_{k-2} and v_k , so C_4 -freeness gives no other; and each v_i with $i \leq k - 3$ has at most one common neighbour with v_k . Hence an extension exists as soon as $d(v_k) \geq k$ (C_4 -free, $k \geq 3$), resp. $d(v_k) \geq k - 1$ and ≥ 2 (girth ≥ 5). With min degree δ the greedy runs to $k = \delta + 1$, resp. $k = \delta + 2$. $\square$

E1 is tight for C_5 and for Petersen. In particular every regular graph of girth ≥ 5 and radius 2 satisfies the conjecture, since there $\ell = \delta$ exactly and $\text{path} \geq \delta + 2 = \text{rad} + \ell$; this covers all Moore graphs of diameter 2.

Theorem 3.7 (geodesic + greedy). *[proved here; assertions machine-checked by wowii133_verify.c]* *If G is connected C_4 -free with $\delta \geq 4$, then $\text{path}(G) \geq \text{rad}(G) + \lfloor \delta/2 \rfloor$.*

Proof. Let c be a centre, $r = \text{rad}$, z at distance r , $u_0 \dots u_r$ a geodesic; set $w_0 = z$ and extend greedily $w_1, w_2, \dots$ at the z end. One step changes the BFS layer from c by at most 1, so w_{j-1} lies in a layer $\geq r - j + 1$, hence $\text{dist}(u_i, w_{j-1}) \geq (r - j + 1) - i$, so only the $\leq j + 2$ vertices u_i with $i \geq r - j - 1$ can be within distance 2 of w_{j-1} . Each path vertex $y \neq w_{j-1}$ blocks at most one candidate, since $|N(y) \cap N(w_{j-1})| \leq 1$, and one further slot is lost to $w_{j-2} \in N(w_{j-1})$. Hence at most $(j + 2) + (j - 2) + 1 = 2j + 1$ neighbours of w_{j-1} are unusable and step j succeeds whenever $d(w_{j-1}) \geq 2j + 2$. All steps $j \leq \lfloor \delta/2 \rfloor - 1$ succeed. $\square$

3.1 Falsification and coverage

Proposition 3.8. *[machine-verified: wowii133_verify.c, wowii133_families.py, wowii133_sa.c]* *No counterexample to Conjecture 133 exists in any of the following ranges:*

range	evaluations	violations
<i>all connected graphs, $n \leq 9$</i>	2 950 981	0
<i>all connected graphs, $n \leq 10$</i>	136 362 861	0
<i>all connected C_4-free graphs, $n \leq 13$</i>	13 544 784	0
<i>79 structured graphs, $n \leq 183$</i>	—	0
<i>simulated annealing, C_4-free, $n = 10..24$</i>	$\approx 2 \cdot 10^6$ proposals	0

The enumerator generates connected graphs by vertex augmentation, which is complete because every connected graph on $k + 1 \geq 2$ vertices has a non-cut vertex; isomorph rejection uses a canonical certificate, and at the last level no rejection is performed, so that level is a superset of all classes. The generator self-validates: its class counts 1, 2, 6, 21, 112, 853, 11117, 261080 for $n = 2..9$ are the numbers of connected graphs on n nodes, and in C_4 -free mode the full enumerator and the C_4 -free-only enumerator agree exactly (36 810 graphs up to $n = 10$). The structured family includes Petersen, the Hoffman–Singleton graph, the incidence graphs of $PG(2, q)$ for $q \leq 9$, the Erdős–Rényi orthogonal polarity graphs ER_q for $q \leq 13$ — the extremal C_4 -free

graphs, and the most dangerous family since they maximise ℓ at radius 2 — cages and random girth-5 regular graphs.

Proposition 3.9 (equality cases). *[machine-verified: `wowii133_verify.c`] Over the whole exhaustive range the C_4 -free branch is tight for exactly five graphs:*

G	n	rad	ℓ	$\lfloor \ell \rfloor$	path	RHS
K_2	2	1	1	1	2	2
K_3	3	1	1	1	2	2
C_5	5	2	2	2	4	4
C_6	6	3	2	2	5	5
<i>Petersen</i>	10	2	3	3	5	5

Consequently no argument that loses even one vertex on these five can prove the conjecture.

Measured against Theorems 3.1, 3.4, 3.6, 3.7 the coverage of the exhaustive C_4 -free range is 100% for $n \leq 10$ and 99.9955% for $n \leq 13$, leaving 606 graphs (516 with rad = 2, 90 with rad = 3, all with $\lfloor \ell \rfloor = 3$), each of them one vertex short. There are 622 C_4 -free graphs with $\lfloor \ell \rfloor = 3$ on $n \leq 13$ vertices at all, and none with $\lfloor \ell \rfloor \geq 4$ below $n = 14$. Theorem 4.5 below closes all 606: all 622 have $\mu \geq 2$, and the residual over the whole exhaustive C_4 -free range becomes 0.

3.2 Auxiliary claims: verified, not proved

The same machinery tested four statements that are *not* used anywhere in this paper and are recorded because one of them is the natural target.

Proposition 3.10. *[machine-verified: `wowii133_verify.c`] The following hold over the ranges stated, with the number of violations shown.*

	<i>claim</i>	<i>range verified</i>	<i>violations</i>
A1	every connected G : path $\geq 2 \text{ rad} - 1$	all connected $n \leq 10$	0
A4	C_4 -free: path $\geq \text{rad} + \delta - 1$	all C_4 -free $n \leq 13$	0
A6	C_4 -free, $G \neq K_3$: path $\geq \text{rad} + \delta$	all C_4 -free $n \leq 13$	1 (K_3)
A7	C_4 -free: $\ell(G) \leq \text{path}(G) - \text{rad}(G)$	all C_4 -free $n \leq 13$; all connected $n \leq 10$	0

None of these is proved. A7 is the floor-free strengthening of (C133-b), i.e. $2m - 3T \leq n(\text{path} - \text{rad})$; it is tight on the same five graphs as Proposition 3.9 and is false without C_4 -freeness ($K_{3,3}$ has $\ell = 3$ and $\text{path} - \text{rad} = 1$).

Theorem 6.7 below proves a non-trivial instance of A7.

4 An endpoint-extension calculus

All lemmas in this section are unconditional facts about C_4 -free graphs. They are the tools that replace the degree-driven bounds of §3, which the surgery obstacle of §9 rules out.

Lemma 4.1 (one-vertex extension). *[proved here; assertions machine-checked by `wowii133_gplus.c`] Let G be C_4 -free with a geodesic $u_0 \dots u_d$, $d \geq 2$, and $a(u_d) \geq 2$. Then $\text{path}(G) \geq d + 2$.*

Proof. By Lemma 2.1 the number of components of $N(u_d)$ is $a(u_d) \geq 2$; take x in a component other than u_{d-1} 's. Then $x \neq u_i$ for all i , since $\text{dist}(u_d, x) = 1$ while $\text{dist}(u_d, u_i) = d - i$, and $i = d - 1$ is excluded; $x \not\sim u_{d-1}$ (different components); $x \not\sim u_{d-2}$, else x, u_d, u_{d-1}, u_{d-2} are four distinct vertices forming a 4-cycle; and for $i \leq d - 3$, $\text{dist}(x, u_i) \geq (d - i) - 1 \geq 2$. So $u_0 \dots u_d x$ is induced. $\square$

Lemma 4.2 (two-vertex extension). *[proved here; assertions machine-checked by wowii133_gplus.c] Let G be C_4 -free with a geodesic $u_0 \dots u_d$, $d \geq 2$, $a(u_0) \geq 3$ and $a(u_d) \geq 2$. Then $\text{path}(G) \geq d + 3$.*

Proof. Take x_1, x_2 in two components of $N(u_0)$ other than u_1 's (possible since $a(u_0) \geq 3$) and y in a component of $N(u_d)$ other than u_{d-1} 's. If y were adjacent to both x_1 and x_2 then u_0 and y would have two common neighbours, a C_4 ; so fix $x \in \{x_1, x_2\}$ with $x \not\sim y$.

We check that $x, u_0, u_1, \dots, u_d, y$ is induced. As in Lemma 4.1, $y \not\sim u_{d-1}$, $y \not\sim u_{d-2}$ and $y \not\sim u_i$ for $i \leq d - 3$, and symmetrically $x \not\sim u_1$, $x \not\sim u_2$ (else u_0, u_2 have the two common neighbours x, u_1), and $x \not\sim u_i$ for $i \geq 3$ by distance. For $d = 2$ the only remaining pair is (y, u_0) : $y \sim u_0$ would give u_0, u_2 the two common neighbours u_1, y . For $d \geq 3$, $\text{dist}(u_0, y) \geq d - 1 \geq 2$. Finally $x \neq y$ and $x \not\sim y$ by choice, and $x, y \notin \{u_0, \dots, u_d\}$ by the distance arguments above. $\square$

Lemma 4.3 (both ends, $d \geq 4$). *[proved here; assertions machine-checked by wowii133_gplus.c] Let G be C_4 -free with a geodesic $u_0 \dots u_d$, $d \geq 4$, $a(u_0) \geq 2$ and $a(u_d) \geq 2$. Then $\text{path}(G) \geq d + 3$. The hypothesis $d \geq 4$ cannot be dropped.*

Proof. Take $x \in N(u_0)$ outside u_1 's component and $y \in N(u_d)$ outside u_{d-1} 's component. Lemma 4.1's checks, applied at both ends, give $x, y \notin \{u_0, \dots, u_d\}$ and non-adjacency of x and of y to every u_i they must avoid. Moreover $\text{dist}(x, y) \geq \text{dist}(u_0, u_d) - 2 = d - 2 \geq 2$, so $x \neq y$ and $x \not\sim y$. Hence $x, u_0, \dots, u_d, y$ is induced. $\square$

For $d = 3$ the statement is false: over the exhaustive C_4 -free range $n \leq 13$ there are 334 pairs (u, z) at distance 3 with both a -values ≥ 2 and $\text{path} < 6$; the two extensions may be adjacent, producing an induced C_6 rather than a P_6 . [machine-verified: wowii133_gplus.c]

Theorem 4.4. *[proved here; assertions machine-checked by wowii133_gplus.c] Every connected C_4 -free graph satisfies $\text{path}(G) \geq \text{rad}(G) + \min\{3, \mu(G)\}$.*

Proof. If $\text{rad} \leq 1$ then $\ell < 2$ by Theorem 3.3, hence $\mu \leq \ell < 2$ so $\mu = 1$, and Theorem 3.1 suffices. Let $\text{rad} \geq 2$. If $\mu = 1$ use Theorem 3.1; if $\mu = 2$ use Theorem 3.4; if $\mu \geq 3$, take a centre c and z at distance rad and apply Lemma 4.2 to a c - z geodesic. $\square$

Theorem 4.4 is tight on Petersen. Two immediate corollaries: a triangle-free C_4 -free graph with $\delta \geq 3$ has $\text{path} \geq \text{rad} + 3$ (there $a = d \geq 3$), and so does any C_4 -free graph with $\delta \geq 5$ (there $a \geq \lceil d/2 \rceil \geq 3$).

The next theorem is the workhorse: it removes the gap at $\mu = 2$, where Theorem 4.4 only gives $\text{rad} + 2$.

Theorem 4.5. *[proved here; assertions machine-checked by wowii133_gplus.c] Let G be connected and C_4 -free with $\mu(G) \geq 2$ and with some vertex h satisfying $a(h) \geq 3$. Then $\text{path}(G) \geq \text{ecc}(h) + 3 \geq \text{rad}(G) + 3$, provided $\text{rad}(G) \geq 2$.*

Proof. Take z with $\text{dist}(h, z) = \text{ecc}(h) =: d \geq \text{rad} \geq 2$. Then $a(h) \geq 3$ and $a(z) \geq \mu \geq 2$, so Lemma 4.2 applies to an h - z geodesic. $\square$

Corollary 4.6. *[proved here; assertions machine-checked by wowii133_gplus.c] Let G be connected and C_4 -free with $\ell(G) \geq 3$ and no vertex of a -value 1 (equivalently, by Lemma 2.2, no leaf and no triangle-leaf). Then $\text{path}(G) \geq \text{rad}(G) + 3$. In particular (C133-b) holds for all such graphs with $\lfloor \ell \rfloor \leq 3$.*

Proof. $\ell \geq 3$ gives $\sum_v a(v) \geq 3n$, so some $a(h) \geq 3$; and $\ell \geq 3 > 2$ with Theorem 3.3 gives $\text{rad} \geq 2$. Apply Theorem 4.5. $\square$

This is what closes the 606 residual graphs of §3.1: all 622 C_4 -free graphs with $\lfloor \ell \rfloor = 3$ on at most 13 vertices have $\mu \geq 2$.

Finally, a lemma that reaches one vertex further at the cost of a hypothesis on a side-neighbour. It is the seed of §8.

Lemma 4.7 (deep endpoint extension, $d \geq 5$). *[proved here; assertions machine-checked by w133_r3_counterexample.py] Let G be C_4 -free with a geodesic $u_0 \dots u_d$, $d \geq 5$, $a(u_0) \geq 3$, $a(u_d) \geq 2$, and a usable side-neighbour x of u_0 with $a(x) \geq 4$. Then $\text{path}(G) \geq d + 4$. In particular $\mu \geq 4$ and $\text{rad} \geq 5$ imply $\text{path} \geq \text{rad} + 4$.*

Proof. Choose x as in the hypothesis and $y \in N(u_d)$ outside u_{d-1} 's component. The graph $N(x)$ has $\geq a(x) \geq 4$ components; u_0 's is one. Since $x \not\sim u_2$ (else u_0, u_2 have common neighbours x, u_1) and $x \not\sim u_3$ (distance), C_4 -freeness gives $|N(x) \cap N(u_2)| \leq 1$ and $|N(x) \cap N(u_3)| \leq 1$, so at most one further component is destroyed by each; take w in a surviving component. Then $w \not\sim u_0$ (different components of $N(x)$); $w \not\sim u_1$, else w, x, u_0, u_1 form a C_4 ; $w \not\sim u_2, u_3$ by the choice of component; and $\text{dist}(w, u_i) \geq i - 2 \geq 2$ for $i \geq 4$. Also $w \notin \{u_0, \dots, u_d\}$ since $\text{dist}(u_0, w) = 2$ and $w \neq u_2$ because $x \not\sim u_2$. Finally $\text{dist}(w, y) \geq (d - 1) - 2 = d - 3 \geq 2$ and $\text{dist}(x, y) \geq (d - 1) - 1 = d - 2 \geq 3$, both using $d \geq 5$. So $w, x, u_0, \dots, u_d, y$ is induced on $d + 4$ vertices. $\square$

The two steps marked “ $d \geq 5$ ” are exactly what fails at $d = 4$ and $d = 3$; §8 repairs the first and §9 shows the second cannot be repaired for free.

5 The radius-2 layer

Write $(\star)$ for the assertion that G has no pair of vertices at distance exactly $\text{rad}(G)$ whose a -values are $(\geq 3, \geq 2)$ in some order. By Lemma 4.2, $(\star)$ is a necessary condition on any graph with $\ell \geq 3$ and $\text{path} < \text{rad} + 3$, and it is the hypothesis under which the following two lemmas run.

Lemma 5.1. *[proved here; assertions machine-checked by w133_res_kill.py] Assume $(\star)$, and let $r = \text{rad}(G)$ in a connected C_4 -free graph. If $a(h) \geq 3$ then $\text{ecc}(h) = r$.*

Proof. Suppose $\text{ecc}(h) \geq r + 1$ and take a geodesic $p_0 = h, p_1, \dots, p_e$ with $e = \text{ecc}(h)$. The vertex p_r is at distance exactly r from h , so $(\star)$ forces $a(p_r) \leq 1$, hence $a(p_r) = 1$; by Lemma 2.2, $N(p_r)$ is a clique, and since it contains p_{r-1} and p_{r+1} we get $p_{r-1} \sim p_{r+1}$. Then $h \rightarrow p_{r-1} \rightarrow p_{r+1}$ is a walk of length r , contradicting $\text{dist}(h, p_{r+1}) = r + 1$. $\square$

Lemma 5.2. *[proved here; assertions machine-checked by w133_res_kill.py] Assume $(\star)$. If $a(h) \geq 3$ and $h = p_0, \dots, p_r = z$ is a geodesic to a vertex at distance $r = \text{rad}$, then $a(p_i) \geq 2$ for every interior vertex $1 \leq i \leq r - 1$.*

Proof. $a(p_i) = 1$ would make $N(p_i)$ a clique containing p_{i-1} and p_{i+1} , hence $p_{i-1} \sim p_{i+1}$ and the geodesic could be shortened. $\square$

So in any graph satisfying $(\star)$ a geodesic from a high vertex to the periphery has the rigid a -profile $(\geq 3), (\geq 2), \dots, (\geq 2), (1)$. The radius-2 case of that rigidity is contradictory.

Theorem 5.3 (the radius-2 layer). *[proved here; assertions machine-checked by w133_res_kill.py] Let G be connected and C_4 -free with $\text{rad}(G) = 2$ and $\ell(G) \geq 3$. Then G has a pair of vertices at distance exactly 2 whose a -values are $(\geq 3, \geq 2)$.*

Proof. Suppose not; then $(\star)$ holds with $r = 2$. From $\ell \geq 3$ we get $\sum_v a(v) \geq 3n$, so some vertex h has $a(h) \geq 3$; by Lemma 5.1, $\text{ecc}(h) = 2$. Put $A := N(h)$, $m := |A|$, and $B := V \setminus (\{h\} \cup A)$, the distance-2 layer. By $(\star)$, every $b \in B$ has $a(b) \leq 1$, hence $a(b) = 1$.

Step 1: unique parents. Every $b \in B$ has at least one neighbour in A (as $\text{dist}(h, b) = 2$) and at most one, since two would give h and b two common neighbours. Write $p(b) \in A$ for that parent and $B_\alpha := p^{-1}(\alpha)$ for the fibre of $\alpha \in A$.

Step 2: no edges between distinct fibres. If $b \in B_\alpha$, $c \in B_\beta$ with $\alpha \neq \beta$ and $b \sim c$, then $N(b)$ contains α and c , which are non-adjacent (the unique parent of c is $\beta \neq \alpha$), so $a(b) \geq 2$, contradicting $a(b) = 1$.

Step 3: the two matchings. $G[A] \subseteq G[N(h)]$ and $G[B_\alpha] \subseteq G[N(\alpha)]$ are matchings plus isolated vertices, by Lemma 2.1. Let p be the number of edges of $G[A]$, and for $\alpha \in A$ let s_α and q_α be the numbers of isolated vertices and of edges of $G[B_\alpha]$; put $S := \sum_\alpha s_\alpha$ and $Q := \sum_\alpha q_\alpha$.

Step 4: the count. First $a(h) = \text{comp}(A) = m - p$. Next fix $\alpha \in A$. By Step 2, $N(\alpha) = \{h\} \cup M(\alpha) \cup B_\alpha$, where $M(\alpha)$ is α 's partner in $G[A]$ if it has one; h is adjacent to $M(\alpha)$ and to nothing in B_α , and no vertex of B_α is adjacent to $M(\alpha)$ (it would then have two parents). Hence the components of $N(\alpha)$ are $\{h\} \cup M(\alpha)$ together with the $s_\alpha + q_\alpha$ components of $G[B_\alpha]$, so $a(\alpha) = 1 + s_\alpha + q_\alpha$. Finally $a(b) = 1$ for $b \in B$ and $|B| = S + 2Q$. Therefore

$$\sum_v a(v) = (m - p) + (m + S + Q) + (S + 2Q) = 2m - p + 2S + 3Q, \quad n = 1 + m + S + 2Q,$$

so that

$$3n - \sum_v a(v) = 3 + m + p + S + 3Q \geq 3 > 0,$$

i.e. $\ell(G) < 3$, contradicting the hypothesis. $\square$

Remark 5.4. The proof never uses any hypothesis on $\text{path}(G)$: the contradiction comes from $(\star)$, $\text{rad} = 2$ and the existence of a high vertex alone. This is what makes Theorem 5.3 composable with Lemma 4.2.

Corollary 5.5 (the radius-2 layer is closed). *[proved here; assertions machine-checked by w133_res_kill.py] Let G be connected and C_4 -free with $\text{rad}(G) = 2$ and $\ell(G) \geq 3$. Then $\text{path}(G) \geq 5 = \text{rad}(G) + 3$. In particular (C133-b) holds for every connected C_4 -free graph with $\text{rad} = 2$ and $|\ell| = 3$.*

Proof. Theorem 5.3 supplies a pair at distance 2 with a -values $(\geq 3, \geq 2)$; apply Lemma 4.2 with $d = 2$. $\square$

6 Peeling induction and the residual problem

From here on the target is

$$(X) \quad G \text{ connected } C_4\text{-free, } \ell(G) \geq 3 \implies \text{path}(G) \geq \text{rad}(G) + 3, \quad (6.1)$$

which implies (C133-b) for $\lfloor \ell \rfloor \leq 3$ and is the whole content of the band $\lfloor \ell \rfloor = 3$. Corollary 4.6 settles (6.1) when there is no vertex of a -value 1; the induction below reduces the general case to a residual that can be attacked.

Lemma 6.1 (deleting an $a = 1$ vertex). *[proved here; assertions machine-checked by `wowii133_plus.c`] Let G be C_4 -free and $a(v) = 1$. Then $G - v$ is connected, C_4 -free, and preserves all distances between the remaining vertices.*

Proof. By Lemma 2.2, v is a leaf or a triangle-leaf. A leaf is never an interior vertex of a shortest path; a triangle-leaf v with $N(v) = \{x, y\}$, $x \sim y$, can always be bypassed by the edge xy . So no shortest path between surviving vertices needs v . $\square$

Lemma 6.2 (peeling arithmetic). *[proved here; assertions machine-checked by `w133_r11_adjudicate.py`] Let G be C_4 -free with $a(v) = 1$, and write $\Sigma(G) := \sum_u a(u)$. Then*

$$\Sigma(G - v) - 3(n - 1) = (\Sigma(G) - 3n) + c, \quad c = \begin{cases} 1 & v \text{ a leaf,} \\ 2 & v \text{ a triangle-leaf.} \end{cases}$$

In particular the slack $\Sigma - 3n$ is strictly increasing under peeling, by at least 1 per deleted vertex.

Proof. If v is a leaf with neighbour x , then v is an isolated vertex of $G[N(x)]$, so $a(x)$ drops by exactly 1 and no other a -value changes: Σ drops by 2 and n by 1, so the slack rises by 1. If v is a triangle-leaf with $N(v) = \{x, y\}$, $x \sim y$, then $\{v, y\}$ is an edge component of $G[N(x)]$ which degenerates to the isolated vertex y ; its contribution to $a(x)$ is still 1, so $a(x)$ is unchanged, and symmetrically $a(y)$ is unchanged. Hence Σ drops by 1 and n by 1, and the slack rises by 2. $\square$

Remark 6.3 (a corrected statement). Earlier internal versions of the project stated Lemma 6.2 as “the slack rises by exactly 1”. That is *false* for triangle-leaves, as the adversarial review of §8 pointed out and as the explicit graph CE-1 of §9 shows: deleting its leaf 1 moves (Σ, n) from $(19, 10)$ to $(17, 9)$, slack $-11 \rightarrow -10$; deleting its triangle-leaf 8 moves it to $(18, 9)$, slack $-11 \rightarrow -9$. Every use in this paper needs only the monotone direction “ $\geq +1$ ”, so no conclusion moves; but the “exactly” was wrong and is corrected here.

Proposition 6.4 (the induction skeleton). *[proved here] Let G be a counterexample to (6.1) of minimum order, and put $\Lambda := \{v : a(v) = 1\}$. Then:*

- (i) $\text{rad}(G) \geq 2$ and $\Lambda \neq \emptyset$;
- (ii) deleting any $v \in \Lambda$ strictly lowers the radius;
- (iii) the core $C := G - \Lambda$ is connected, C_4 -free, isometrically embedded in G , every vertex of C has $a_C \geq 2$, and $\ell(C) > 3$;
- (iv) $\text{diam}(G) \leq \text{rad}(G) + 1$, and $(\star)$ holds in G .

Proof. (i) $\ell \geq 3 > 2$ and Theorem 3.3 give $\text{rad} \geq 2$; $\Lambda = \emptyset$ would contradict Corollary 4.6. (ii) If $\text{rad}(G - v) = \text{rad}(G)$ for some $v \in \Lambda$ then $G - v$ is connected, C_4 -free (Lemma 6.1) with $\ell \geq 3$ (Lemma 6.2) and smaller order, so by minimality $\text{path}(G - v) \geq \text{rad}(G - v) + 3 = \text{rad}(G) + 3$, and $\text{path}(G) \geq \text{path}(G - v)$, a contradiction. (iii) Connectivity and isometry are Lemma 6.1 applied to the whole of Λ at once, whose proof shows directly that no shortest path between core vertices uses a vertex of Λ . Deleting the members of Λ one at a time — each remains of a -value 1 when its turn comes, a triangle-leaf possibly having become a leaf — and applying Lemma 6.2 $|\Lambda| \geq 1$ times gives $\Sigma(C) - 3n_C \geq (\Sigma(G) - 3n) + |\Lambda| \geq 1$, i.e. $\ell(C) > 3$. (iv) If $\text{diam} \geq \text{rad} + 2$ a diametral geodesic is an induced path on $\text{diam} + 1 \geq \text{rad} + 3$ vertices. $(\star)$ is Lemma 4.2 together with $\text{path}(G) \leq \text{rad}(G) + 2$. $\square$

For $\text{rad} \geq 4$ one obtains in addition, from Lemma 4.3, that all vertices of a -value ≥ 2 lie pairwise within distance $\text{rad} - 1$, whence $\text{diam}(C) \leq \text{rad} - 1$ and C is self-centred with $\text{rad}(C) = \text{diam}(C) = \text{rad}(G) - 1 =: D$; moreover every core vertex lies at distance exactly D from some vertex of the “parent set” $P := \{\text{core vertices with a } \Lambda\text{-neighbour}\}$. Applying minimality to C gives $\text{path}(C) \geq D + 3 = \text{rad}(G) + 2$, one vertex short. This is the residual:

Problem 6.5 (the residual problem). Let C be connected, C_4 -free, self-centred with $\text{rad}(C) = \text{diam}(C) = D$ and $\ell(C) > 3$, and let $P \subseteq V(C)$ satisfy: for every u there is $p \in P$ with $\text{dist}(u, p) = D$. Show that C has an induced path on $D + 3$ vertices with at least one end in P (or one on $D + 2$ vertices with both ends in P).

Problem 6.6 (the strengthened form). C connected C_4 -free, $\text{diam}(C) = D$, $\ell(C) > 3 \implies \text{path}(C) \geq D + 4$.

Problem 6.6 implies Problem 6.5, since $D + 4 > D + 3$ makes the endpoint condition free, and it is the integer form of the strengthening A7 of Proposition 3.10 in the self-centred case. Its case $D = 2$ is the following theorem; $D \geq 5$ is available from Lemma 4.7 under an extra side-neighbour hypothesis, $D = 4$ from Lemma 8.1 below, and $D = 3$ is genuinely open — indeed §9 shows that the local route to it is closed.

Theorem 6.7 ($\text{diam} = 2$). *[proved here; assertions machine-checked by `wowii133_gplus.c`, `w133_g12_nonempty.py`] Let C be connected, C_4 -free, with $\text{diam}(C) = 2$ and $\ell(C) > 3$ (strictly). Then $\text{path}(C) \geq 6$. Since $\text{diam} = 2$ and $\ell > 2$ force $\text{rad}(C) = 2$ by Theorem 3.3, this says $\text{path} \geq \text{rad} + 4$.*

Proof. Step 0. $\text{diam} = 2$ and C_4 -freeness make any two non-adjacent vertices have *exactly* one common neighbour.

Step 1: the triangle-free case. We first record the step that an adversarial review found missing in an earlier version of this proof.

Claim. If C is connected, C_4 -free, triangle-free with $\text{diam}(C) = 2$ and $\ell(C) > 3$, then C contains a cycle and its girth is exactly 5.

Proof. If C were acyclic it would be a tree of diameter 2, i.e. a star $K_{1,m}$, and $\ell(K_{1,m}) = 2m/(m+1) < 2$, contradicting $\ell > 3$. A graph containing a cycle satisfies $\text{girth} \leq 2 \text{diam} + 1 = 5$, and triangle-freeness with C_4 -freeness gives $\text{girth} \geq 5$.

Without this step the claim “girth 5” is false: a star is triangle-free, C_4 -free and of diameter 2 but has no cycle at all, and is not regular. Given girth 5 and diameter 2, C is a Moore graph,

hence k -regular with $k \in \{2, 3, 7, 57\}$; being triangle-free, $\ell(C) = k$, so $\ell > 3$ forces $k \in \{7, 57\}$ and $\delta \geq 7$, and Theorem 3.6(E1) gives $\text{path} \geq \delta + 2 \geq 9 \geq 6$.

Step 2: the case with a triangle. Suppose $\text{path}(C) \leq 5$ and let $\{a, b, c\}$ be a triangle. Put $A := N(a) \setminus \{b, c\}$, $B := N(b) \setminus \{a, c\}$, $C_s := N(c) \setminus \{a, b\}$ and $R := V \setminus (\{a, b, c\} \cup A \cup B \cup C_s)$.

(i) A, B, C_s are pairwise disjoint and pairwise non-adjacent. $x \in A \cap B$ would make c, x two common neighbours of a, b ; and $\alpha \in A$, $\beta \in B$ with $\alpha \sim \beta$ would make b, α two common neighbours of a, β .

(ii) Every $u \in R$ has exactly one neighbour in each of A, B, C_s . As $u \not\sim a$ and $\text{diam} = 2$, u and a have exactly one common neighbour $w \in \{b, c\} \cup A$; since $u \not\sim b, c$ we get $w \in A$.

(iii) *Counting identities.* $a(a) = 1 + \text{comp}(A)$, and for $\alpha \in A$, $a(\alpha) = 1 + \text{comp}(R_\alpha)$ where $R_\alpha := N(\alpha) \cap R$. Indeed $N(a) = \{b, c\} \cup A$ whose internal edges are bc together with the edges of A (no b - A edges by (i)); and $N(\alpha) = \{a\} \cup M(\alpha) \cup R_\alpha$ with internal edges a - $M(\alpha)$ and those of R_α , since $M(\alpha) \subseteq A$ has no R_α -neighbour by (ii).

(iv) R is a clique. Let $\rho, \sigma \in R$ be non-adjacent, and let f_A, f_B, f_C denote the unique neighbours supplied by (ii). If $f_A(\rho) \neq f_A(\sigma)$ and $f_B(\rho) \neq f_B(\sigma)$, put $\alpha := f_A(\rho)$, $\beta := f_B(\sigma)$; then $\rho, \alpha, a, b, \beta, \sigma$ is an induced P_6 — $\rho \not\sim a, b$ by the definition of R ; $\rho \not\sim \beta$ since ρ 's unique B -neighbour differs from β ; $\alpha \not\sim b, \beta, \sigma$ and $a \not\sim \beta, \sigma$, $b \not\sim \sigma$ by (i),(ii) — contradicting $\text{path} \leq 5$. Using the edges ac and bc likewise, at least two of the three coordinates coincide; but two coincidences give ρ, σ two common neighbours, a C_4 .

(v) $|R| \leq 3$, since a clique on 4 vertices contains a C_4 .

(vi) *The count.* Write $s := |A| + |B| + |C_s|$ and $\rho := |R|$. By (iii), $\sum_v a(v) = 3 + \sum_T \text{comp} + s + \sum_\alpha \text{comp}(R_\alpha) + \sum_{u \in R} a(u)$, where $\sum_T \text{comp} \leq s$ and $\sum_\alpha \text{comp}(R_\alpha) \leq 3\rho$ (each R -vertex lies in exactly one fibre on each of the three sides). Now $n = 3 + s + \rho$, and we compare with $3n$:

- $\rho = 0$: $a(\alpha) = 1$ for all α , so $\sum a \leq 3 + 2s$ while $3n = 9 + 3s$.
- $\rho = 1$: $d(u) = 3$ so $a(u) \leq 3$, $\sum a \leq 9 + 2s$ while $3n = 12 + 3s$.
- $\rho = 2$: $d(u) = 4$ so $a(u) \leq 4$, $\sum a \leq 17 + 2s$ while $3n = 15 + 3s$, forcing $s < 2$; but (ii) gives $s \geq 3$.
- $\rho = 3$: R is a triangle, and for $u \in R$, $N(u) = \{\alpha_u, \beta_u, \gamma_u\} \cup (R \setminus \{u\})$ where $R \setminus \{u\}$ is an edge; by the matching property $\alpha_u, \beta_u, \gamma_u$ are non-adjacent to $R \setminus \{u\}$, so $a(u) = 4$ exactly, and f_A is injective on R , giving $|A|, |B|, |C_s| \geq 3$ and $s \geq 9$. Then $\sum a \leq 3 + 2s + 9 + 12 = 24 + 2s$ while $3n = 18 + 3s$, forcing $s < 6$.

In every case $\sum_v a(v) \leq 3n$, contradicting $\ell(C) > 3$. $\square$

Remark 6.8 (status of Theorem 6.7). Step 1's Claim was added after an adversarial review pass found the gap; the star counterexample above is that reviewer's. The patched statement has *not* itself been through a second, cross-family review pass — that item is still queued in the campaign's ledger — which is why Theorem 6.7 carries [proved here; assertions machine-checked by] and not [proved here; machine-checked by ; **two-family review**]. The structural steps (i)–(iv) were asserted over the exhaustive C_4 -free range $n \leq 13$ on 422 non-vacuous diameter-2 instances with 0 failures; the theorem's own hypothesis class is empty in that range (the closest graph, Petersen, has $\ell = 3$ exactly and is excluded by strictness), so its positive evidence is the explicit witness below.

Proposition 6.9 (non-vacuity of Theorem 6.7). *[machine-verified: w133_g12_nonempty.py, w133_res3_search.py] Both branches of Theorem 6.7's hypothesis class are inhabited. The Hoffman–Singleton graph has $n = 50$, girth 5, $\text{diam} = 2$, is 7-regular and triangle-free, so $\ell = 7 > 3$; it satisfies the conclusion with an induced-path certificate of 20 vertices. The Erdős–Rényi polarity graphs ER_5, ER_7, ER_{11} are C_4 -free of diameter 2 with triangles and $\ell = 3.871, 4.912, 6.947$ respectively, with induced-path certificates 15, 22, 37.*

7 The radius-3 layer

This section proves the main theorem of the paper.

Theorem 7.1 (the radius-3 layer). *[proved here] Let G be a counterexample to (6.1) of minimum order. Then $\text{rad}(G) \geq 4$.*

The proof occupies §7.1–§7.3. It reduces the case $\text{rad}(G) = 3$ to the emptiness of an explicit class, then proves that emptiness in two theorems.

7.1 The reduction

Definition 7.2. Write (R3b) for the property: *every vertex of a -value ≥ 3 has eccentricity ≤ 2* , and let

$$\text{RES} := \{ C : C \text{ connected, } C_4\text{-free, } \text{rad}(C) = 2, \text{diam}(C) = 3, \ell(C) > 3, \text{ (R3b)} \}.$$

Write (C6) for the property that C has no induced P_6 , i.e. $\text{path}(C) \leq 5$.

Two remarks on the shape of this definition, both of which cost the campaign a round and are recorded so that the reader does not repeat them.

Remark 7.3 (the hypothesis (R3b) is not optional). *[machine-verified: w133_res3_search.py]* The class $\{C_4\text{-free, rad} = 2, \text{diam} = 3, \ell > 3\}$ — i.e. RES without (R3b) — is *inhabited*, with at least nine explicit members: Hoffman–Singleton minus 1, 2, 3, 5, 8 or 12 vertices (ℓ from 6.857 down to 5.368) and ER_5, ER_7 minus one vertex ($\ell = 3.900, 4.875$). All satisfy the conjecture, with induced-path certificates ≥ 15 . So no contradiction can be derived from that class as stated. Every one of the nine violates (R3b), for a structural reason: once a dense core has diameter 3, the pair realising the diameter consists of high vertices, which (R3b) forbids.

Remark 7.4 (which reading of (R3)). *[machine-verified: w133_res_kill.py]* The weaker reading (R3a) — *no pair at distance ≥ 3 with a -values $(\geq 3, \geq 2)$* — is *not* equivalent to (R3b), and the class it defines is inhabited: Hoffman–Singleton with one pendant vertex, with one triangle-leaf, and ER_q with one pendant vertex ($q = 5, 7, 11$) all satisfy (R3a) and fail (R3b), because the unique vertex at distance 3 from the offending high vertex is the grafted $a = 1$ vertex, which (R3a) never sees. All proofs below use (R3b), which is available only through Lemma 7.5.

Lemma 7.5 (core lifting). *[proved here] Let G be a minimum-order counterexample to (6.1) with $\text{rad}(G) = 3$, let $\Lambda = \{v : a_G(v) = 1\}$ and $C = G - \Lambda$. Then (R3b) holds in C .*

Proof. By Proposition 6.4(iv), G satisfies $(\star)$ at $r = 3$; more precisely, $\text{path}(G) \leq \text{rad}(G) + 2 = 5$ and Lemma 4.2 give: no pair of G at distance ≥ 3 has a_G -values $(\geq 3, \geq 2)$. Every vertex of C has $a_G \geq 2$ by the definition of Λ . Let $h \in C$ with $a_C(h) \geq 3$; deleting vertices cannot increase a -values, so $a_G(h) \geq 3$. If $\text{ecc}_C(h) = 3$, pick $z \in C$ with $\text{dist}_C(h, z) = 3 = \text{dist}_G(h, z)$ by isometry (Proposition 6.4(iii)); then $a_G(z) \geq 2$ and (h, z) is a forbidden pair. $\square$

Proposition 7.6 (the reduction). *[proved here] Let G be a minimum-order counterexample to (6.1) with $\text{rad}(G) = 3$. Then the core $C = G - \Lambda$ belongs to RES and satisfies (C6).*

Proof. By Proposition 6.4, C is connected, C_4 -free, isometric in G , with $\ell(C) > 3$ and all a_G -values ≥ 2 . If C contained a pair at distance ≥ 4 , both its ends would have $a_C \geq 1$; more usefully, both have $a_G \geq 2$ and hence, by isometry and Lemma 4.3, $\text{path}(G) \geq 4 + 3 = 7 > 5$. So $\text{diam}(C) \leq 3$. If $\text{diam}(C) = 2$, Theorem 6.7 gives $\text{path}(G) \geq \text{path}(C) \geq 6$, a contradiction. So $\text{diam}(C) = 3$. If $\text{rad}(C) = 3$, minimality applied to C (which has $\ell(C) > 3$ and fewer vertices) gives $\text{path}(C) \geq \text{rad}(C) + 3 = 6$, again a contradiction; so $\text{rad}(C) = 2$. Property (R3b) is Lemma 7.5, and $\text{path}(C) \leq \text{path}(G) \leq 5$ is (C6). $\square$

Theorem 7.1 therefore follows from Theorems 7.13 and 7.23 below, which together say $\text{RES} \wedge (C6) = \emptyset$: with Corollary 5.5 disposing of $\text{rad} = 2$ and Proposition 6.4(i) of $\text{rad} \leq 1$, a minimum-order counterexample must have radius at least 4.

Throughout §7.1–§7.3 we fix $C \in \text{RES}$, $H := \{v : a(v) \geq 3\}$ (all a -values are computed in C), and we use the following mass inequality repeatedly.

Lemma 7.7 (the mass inequalities). *[proved here; assertions machine-checked by `w133_res_kill.py`] Let C be C_4 -free with $\ell(C) > 3$.*

- (a) *If every vertex outside H has $a \leq 2$ — which holds by definition of H — then $\sum_{h \in H} (a(h) - 2) > n$. In particular $H \neq \emptyset$.*
- (b) *If c has $\text{ecc}(c) = 2$, $A := N(c)$, $k := |A|$ and B is the distance-2 layer, then $\sum_{b \in B} a(b) > 3 + k + 2|B|$; consequently B contains a vertex of H .*

Proof. (a) $3n < \sum_v a(v) \leq \sum_H a(h) + 2(n - |H|)$. (b) As in the proof of Theorem 5.3, every $b \in B$ has a unique parent in A , $a(c) = \text{comp}(A) \leq k$, and $a(\alpha) = 1 + \text{comp}(B_\alpha)$ for $\alpha \in A$, with $\sum_\alpha \text{comp}(B_\alpha) \leq |B|$. Hence $\sum_v a(v) \leq k + (k + |B|) + \sum_B a(b)$, while $\sum_v a(v) > 3n = 3(1 + k + |B|)$. The consequence follows since $a(b) \leq 2$ for all $b \in B$ would give $\sum_B a(b) \leq 2|B|$. $\square$

Lemma 7.8 (mass–Bonferroni). *[proved here; assertions machine-checked by `w133_res_kill.py`] In any C_4 -free graph and any $S \subseteq V$, $\sum_{h \in S} a(h) \leq |N(S)| + P(S) \leq n + \binom{|S|}{2}$, where $N(S) = \bigcup_{h \in S} N(h)$ and $P(S)$ is the number of pairs of S with a common neighbour. Consequently, in RES, $|H| \geq 6$.*

Proof. Write $a(h) = |I_h|$ for a maximum independent set $I_h \subseteq N(h)$, and $m(v) := |\{h \in S : v \in I_h\}| \leq \deg_S(v)$. Then $\sum_S a = \sum_v m(v) \leq |N(S)| + \sum_v (m(v) - 1)^+ \leq |N(S)| + \sum_v \binom{\deg_S(v)}{2}$, and the last sum counts pairs of S with a common neighbour, with multiplicity equal to the number of common neighbours, which is ≤ 1 by C_4 -freeness. For the consequence: Lemma 7.7(a) gives $\sum_H a > n + 2|H|$, so $n + \binom{|H|}{2} > n + 2|H|$, i.e. $|H| \geq 6$. $\square$

Lemma 7.9 (no leaf at eccentricity 3). *[proved here; assertions machine-checked by `w133_res_kill.py`] In RES no vertex of eccentricity 3 is a leaf.*

Proof. Let u be a leaf with neighbour p and $\text{ecc}(u) = 3$. Then $\text{dist}(u, \cdot) = 1 + \text{dist}(p, \cdot) \leq 3$ gives $\text{ecc}(p) \leq 2$, and $\text{ecc}(p) = 1$ would force $\text{diam} \leq 2$; so $\text{ecc}(p) = 2$. By (R3b) every $h \in H$ has $\text{dist}(h, u) \leq 2$, hence $\text{dist}(h, p) \leq 1$, i.e. no vertex of H lies at distance exactly 2 from p — contradicting Lemma 7.7(b) at $c = p$. $\square$

7.2 The triangular endpoint branch is empty

By (R3b), a pair (u, z) realising $\text{diam}(C) = 3$ has $a(u), a(z) \leq 2$. This subsection kills the case $a(u) = 1$; §7.3 kills $a(u) = a(z) = 2$.

By Lemma 2.2 and Lemma 7.9, a vertex u with $a(u) = 1$ and $\text{ecc}(u) = 3$ has $d(u) = 2$ and $N(u) = \{p, q\}$ with $p \sim q$: a *triangular endpoint*. Fix such a u and write

$$B_p := N(p) \setminus \{u, q\}, \quad B_q := N(q) \setminus \{u, p\}, \quad D := \text{the distance-3 layer from } u,$$

$\beta_x := |B_x|$, $\delta_h := |N(h) \cap D|$, $H' := H \cap (B_p \cup B_q)$, $\eta_x := |H \cap B_x|$, $e_x :=$ the number of edges inside B_x .

Lemma 7.10. *[proved here; assertions machine-checked by w133_r7_triend.py] With this notation:*

- (a) $N(p) \cap N(q) = \{u\}$;
- (b) $\text{ball}_2(u) = \{u, p, q\} \sqcup B_p \sqcup B_q$ and $|\text{ball}_2(u)| = d(p) + d(q) - 1$; in particular $n = 3 + \beta_p + \beta_q + |D|$ and neither p nor q has a neighbour in D ;
- (c) no vertex of B_p is adjacent to q , every vertex has at most one neighbour in B_p and at most one in B_q ;
- (d) $a(p) - 2 = \beta_p - 1 - e_p$ and $a(q) - 2 = \beta_q - 1 - e_q$;
- (e) $H \subseteq \{p, q\} \cup B_p \cup B_q$.

Proof. (a) $N(u) = \{p, q\}$ is a clique, and two common neighbours of p, q besides u would give a C_4 . (b) Every $b \in \text{ball}_2(u) \setminus \{u, p, q\}$ has exactly one neighbour in $\{p, q\}$, at least one by definition and at most one by (a); the count is $3 + (d(p) - 2) + (d(q) - 2)$. A neighbour of p in D would be at distance 2 from u . (c) $h \in B_p$ adjacent to q would lie in $N(p) \cap N(q) = \{u\}$; two neighbours of a vertex w inside B_p would give w and p two common neighbours. (d) $N(p) = \{u, q\} \cup B_p$ and its internal edges are uq together with the e_p edges of B_p : there are no q - B_p edges by (c) and no u - B_p edges since $N(u) = \{p, q\}$. So $t(p) = 1 + e_p$, $d(p) = 2 + \beta_p$ and Lemma 2.1 gives $a(p) = 1 + \beta_p - e_p$. (e) By (R3b) every $h \in H$ has $\text{ecc}(h) \leq 2$, so $H \subseteq \text{ball}_2(u)$; and $u \notin H$. $\square$

Note that (d) forces $a(p) \geq 2$ as soon as $\beta_p \geq 1$, because $e_p \leq \lfloor \beta_p/2 \rfloor \leq \beta_p - 1$ for $\beta_p \geq 2$ and $e_p = 0$ for $\beta_p = 1$. This is used below to add the p - and q -terms to the mass sum whether or not they are hubs.

Lemma 7.11 (P_k -endpoint degree bound). *[proved here; assertions machine-checked by w133_r6_adjudicate.py] Let G be C_4 -free with no induced P_{k+1} , $k \geq 3$. Then every endpoint of an induced P_k has degree $\leq k$. In particular, C_4 -free with no induced P_6 implies every endpoint of an induced P_5 has degree ≤ 5 .*

Proof. Let $(v_0, \dots, v_{k-1})$ be an induced P_k and $w \in N(v_0) \setminus \{v_1, \dots, v_{k-1}\}$. If w had no neighbour among $v_1, \dots, v_{k-1}$ then $(w, v_0, \dots, v_{k-1})$ would be an induced P_{k+1} . So $w \in N(v_0) \cap N(v_i)$ for some $i \geq 1$, and each such intersection has size ≤ 1 by C_4 -freeness: that is at most $k-1$ vertices, and adding v_1 gives $d(v_0) \leq k$. $\square$

The hypothesis “endpoint of an induced P_5 ” cannot be dropped: the friendship graphs F_k are C_4 -free and P_6 -free with unbounded degree $2k$, and contain no induced P_5 at all. [machine-verified: w133_r6_adjudicate.py]

Lemma 7.12. *[proved here; machine-checked by w133_r7_friend.py; two-family review] In RES with a triangular endpoint u :*

- (a) *there is no edge between B_p and B_q ;*
- (b) *every $h \in B_p \cup B_q$ satisfies $a(h) \leq 1 + \delta_h$;*
- (c) *if $b \in B_p$ has $\text{ecc}(b) = 2$ then $\beta_q \leq \delta_b$, and symmetrically;*
- (d) *$\beta_p, \beta_q \geq 1$;*
- (e) *under (C6), every hub of $B_p \cup B_q$ has degree ≤ 5 , hence $\delta_b \leq 4$ for such a hub; and with (c), $\beta_p, \beta_q \leq 4$ as soon as both sides carry a hub.*

Proof. (a) If $h \in B_p$, $c \in B_q$ and $h \sim c$, then p and c have the two common neighbours h and q (using $p \sim q$).

(b) $|N(h) \cap B_p| \leq 1$ by Lemma 7.10(c). If $N(h) \cap B_p = \{b\}$ then pb is an edge inside $N(h)$, so $a(h) \leq d(h) - 1 = 1 + \delta_h$ using (a); otherwise $d(h) = 1 + \delta_h \geq a(h)$.

(c) For $c \in B_q$ we have $\text{dist}(b, c) \geq 2$ by (a), so $\text{ecc}(b) = 2$ gives a common neighbour w of b and c . Now $w \neq p$ (as $p \not\sim c$ by Lemma 7.10(a)), $w \notin B_p$ by (a), and $w \neq u, q$; so $w \in N(b) \cap D$. Each D -vertex has at most one neighbour in B_q , so $c \mapsto w$ is injective.

(d) Suppose $B_q = \emptyset$. Then $N(q) = \{u, p\}$ with $u \sim p$, so $a(q) = 1$ and $q \notin H$; every $y \in D$ has at most one neighbour in $B_p \cup B_q = B_p$. From $a(p) \leq d(p) - 1 = \beta_p + 1$ and $a(h) - 2 \leq \delta_h$ for $h \in H'$ (which is (b) weakened), Lemma 7.7(a) gives

$$n < \sum_{h \in H} (a(h) - 2) \leq (\beta_p - 1) + \sum_{h \in H'} \delta_h \leq (\beta_p - 1) + |D| = (n - 3) - 1 = n - 4,$$

a contradiction.

(e) Let $h \in H \cap B_p$ with $d(h) \geq 6$. By (R3b), $\text{ecc}(h) \leq 2$, and $\text{ecc}(h) = 1$ would force $\text{diam} \leq 2$; so $\text{ecc}(h) = 2$. By (d), $\beta_q \geq 1$. If some $y \in N(h) \cap D$ had a neighbour $c \in B_q$, then $c \notin N(h)$ by (a) and (h, y, c, q, u) is an induced P_5 — $h \not\sim c$, $h \not\sim q$, $h \not\sim u$, $y \not\sim q, u$ (as $y \in D$), $c \not\sim u$ — so Lemma 7.11 gives $d(h) \leq 5$, a contradiction. Hence no vertex of $N(h) \cap D$ has a B_q -neighbour and the injection of (c) has empty target, giving $\beta_q \leq 0$: contradiction. The last clause is (c) together with $\delta_b \leq d(b) - 1 \leq 4$. $\square$

Theorem 7.13 (the triangular branch is empty). *[proved here; machine-checked by w133_r7_friend.py; two-family review] In $\text{RES} \wedge (C6)$ no vertex has a -value 1 and eccentricity 3. Equivalently, every pair of vertices at distance 3 has both a -values equal to 2.*

Proof. Suppose u is such a vertex and keep the notation above. Let $D_2 := \{y \in D : y \text{ has a neighbour in } H' \cap B_p \text{ and one in } H' \cap B_q\}$.

Step 1: a lower bound on $|D_2|$. By Lemma 7.10(d),(e) and Lemma 7.12(b), and since $a(p) - 2, a(q) - 2 \geq 0$ by Lemma 7.12(d),

$$n < \sum_{h \in H} (a(h) - 2) \leq (\beta_p - 1 - e_p) + (\beta_q - 1 - e_q) + \sum_{h \in H'} (\delta_h - 1).$$

Each $y \in D$ has at most one neighbour in B_p and one in B_q , so it contributes at most 1 to $\sum_{h \in H'} \delta_h$, and 2 exactly when $y \in D_2$; hence $\sum_{h \in H'} \delta_h = \sum_{y \in D} |N(y) \cap H'| \leq |D| + |D_2|$. Substituting $n = 3 + \beta_p + \beta_q + |D|$ and $|H'| = \eta_p + \eta_q$ yields

$$|D_2| \geq 6 + \eta_p + \eta_q + e_p + e_q.$$

Two vertices of D_2 with the same pair of parents would be two common neighbours of that pair, a C_4 ; so $y \mapsto (\text{its } B_p\text{-parent, its } B_q\text{-parent})$ is injective and $|D_2| \leq \eta_p \eta_q$. Therefore

$$(\eta_p - 1)(\eta_q - 1) \geq 7 + e_p + e_q.$$

Step 2: both sides are saturated. The displayed inequality forces $\eta_p, \eta_q \geq 2$, so both sides carry hubs and Lemma 7.12(e) gives $\beta_p, \beta_q \leq 4$, hence $\eta_p, \eta_q \leq 4$. The only integer pair with both entries ≤ 4 and $(\eta_p - 1)(\eta_q - 1) \geq 7$ is $\eta_p = \eta_q = 4$; so $\beta_p = \beta_q = 4$, every vertex of $B_p \cup B_q$ is a hub, and $d(p) = d(q) = 6$.

Step 3: the B -vertices are pinned. For a hub $b \in B_p$, Lemma 7.12(c) gives $\delta_b \geq \beta_q = 4$ while Lemma 7.12(e) gives $5 \geq d(b) = 1 + |N(b) \cap B_p| + \delta_b$. Hence $|N(b) \cap B_p| = 0$, $\delta_b = 4$, $d(b) = 5$; therefore $e_p = e_q = 0$ and $a(p) = a(q) = 5$.

Step 4: the graph is determined. $a(p) = 5 \geq 3$ makes p a hub, so $\text{ecc}(p) = 2$ and every $y \in D$ has a B_p -parent, and symmetrically a B_q -parent. Counting B_p - D edges gives $|D| = \sum_{b \in B_p} \delta_b = 16$, and injectivity makes $y \mapsto (b, c)$ a bijection $D \rightarrow B_p \times B_q$. Thus $n = 3 + 4 + 4 + 16 = 27$.

Step 5: the contradiction. Fix $y \in D$ with B_p -parent b_1 and let b be one of the other three vertices of B_p . As b is a hub, $\text{ecc}(b) = 2$, so b and y have a common neighbour w ; $w \neq b_1$ since $e_p = 0$, and $w \notin B_q$ by Lemma 7.12(a), so $w \in N(y) \cap D$ — and the unique B_p -parent of w is b , so the three choices of b give three distinct w . Hence $d(y) \geq 2 + 3 = 5$ and, in a C_4 -free graph, $a(y) \geq \lceil d(y)/2 \rceil \geq 3$: y is a hub at distance 3 from u , contradicting $H \subseteq \text{ball}_2(u)$. $\square$

Corollary 7.14. *[proved here; machine-checked by `w133_r7_triend.py`; **two-family review**] In $\text{RES} \wedge (C6)$ every pair (u, z) at distance 3 satisfies $a(u) = a(z) = 2$.*

Remark 7.15. The proof of Theorem 7.13 never uses the branch hypothesis that every diameter pair has a low endpoint — only the existence of one vertex of a -value 1 at eccentricity 3 — which is why it yields the stronger corollary. Property (C6) enters only through Lemma 7.12(e), via Lemma 7.11. No exhaustive search over graphs is involved: the only sweeps are the small-graph table backing the unconditional lemmas and two bounded integer sweeps over $(\eta_p, \eta_q, e_p, e_q) \in [0, 4]^4$.

7.3 The hexagon branch is empty

By Corollary 7.14 the diameter pair (u, z) of $C \in \text{RES} \wedge (C6)$ has $a(u) = a(z) = 2$. The next lemma turns that into an induced hexagon.

Lemma 7.16 (the forced hexagon). *[proved here; assertions machine-checked by `wowii133_gplus.c`] Let G be C_4 -free with $\text{dist}(u, z) = 3$ and $a(u) = a(z) = 2$. Then either $\text{path}(G) \geq 6$, or: $N(u)$ consists of u_1 's component together with a singleton component $\{x\}$, $N(z)$ consists of u_2 's component together with a singleton $\{y\}$, and $x \sim y$ — so that G contains an induced 6-cycle through u and z .*

Proof. Fix a geodesic u, u_1, u_2, z . Take $x \in N(u)$ outside u_1 's component and $y \in N(z)$ outside u_2 's component. By the checks in Lemma 4.1 applied at both ends, $x, y \notin \{u, u_1, u_2, z\}$ and $x \not\sim u_1, u_2, z$, $y \not\sim u_2, u_1, u$. If some legitimate choice has $x \not\sim y$, then x, u, u_1, u_2, z, y is an induced P_6 . So suppose every legitimate x is adjacent to every legitimate y . If x 's component had two vertices $x, \bar{x}$, then x and $\bar{x}$ would both be adjacent to u and to y , a C_4 ; so it is a singleton, and symmetrically for y . $\square$

So C carries an induced hexagon $Z = (u, u_1, u_2, z, y, x)$ at cyclic positions 0..5, with $\{x\}$ a singleton component of $N(u)$ and $\{y\}$ a singleton component of $N(z)$. The next lemma is the engine of the whole collapse and is of independent interest.

Lemma 7.17. *[proved here; machine-checked by w133_r8_q61.py; two-family review] Let $Z = (v_0, \dots, v_5)$ be an induced C_6 in a graph with no induced P_6 . Then no vertex outside Z has exactly two Z -neighbours that are consecutive on Z . (No C_4 -freeness and no other hypothesis is used.)*

Proof. Suppose $w \notin Z$ has exactly the two Z -neighbours v_i, v_{i+1} . Deleting v_{i+1} from Z leaves the induced path $(v_i, v_{i-1}, v_{i-2}, v_{i-3}, v_{i-4})$ on the remaining five vertices, with endpoints v_i and $v_{i-4} = v_{i+2}$; and w is adjacent to v_i only among them, its other Z -neighbour having been deleted. Hence $(w, v_i, v_{i-1}, v_{i-2}, v_{i-3}, v_{i-4})$ is an induced P_6 . $\square$

Lemma 7.18 (antipodal rigidity). *[proved here; machine-checked by w133_r8_q61.py; two-family review] Let $Z = (v_0, \dots, v_5)$ be an induced C_6 in a C_4 -free graph with no induced P_6 . Then every vertex outside Z has 0 or 2 neighbours on Z , and if 2 they are antipodal (cyclic distance exactly 3).*

Proof. Let $w \notin Z$.

$|N(w) \cap Z| \neq 1$: if v_i is the unique Z -neighbour, delete v_{i+1} ; the resulting induced P_5 has v_i as an endpoint and w extends it to an induced P_6 .

$|N(w) \cap Z| \leq 2$: any three vertices of a hexagon contain a pair at cyclic distance 2. Indeed three vertices split the cycle into arcs of lengths $a+b+c=6$ with $a, b, c \geq 1$, and a pair separated by an arc of length j is at cyclic distance $\min(j, 6-j)$, which equals 2 exactly when $j \in \{2, 4\}$; if no pair were at cyclic distance 2 we would need $a, b, c \in \{1, 3, 5\}$ with $a+b+c=6$, which has no solution. For such a pair v_j, v_{j+2} , the vertices w and v_{j+1} have the two common neighbours v_j, v_{j+2} , a C_4 .

Cyclic distance 2 is excluded by the same C_4 , and cyclic distance 1 by Lemma 7.17. $\square$

Remark 7.19 (a statement that outran its proof). [machine-verified: w133_r8_judge20_g32.py] Lemma 7.18 is *false* without C_4 -freeness: C_6 together with a vertex w adjacent to v_0 and v_2 is P_6 -free, and w has two Z -neighbours at cyclic distance 2. The first version of this lemma in the campaign omitted the hypothesis and was refuted by the blind reviewer; the genuinely hypothesis-free half is Lemma 7.17. The defect was inherited: an upstream lemma about induced hexagons had been *stated* with the conclusion “0 or 2 neighbours” while its written proof only excluded “exactly 1”, never “ ≥ 3 ”, and the present Lemma 7.18 copied the half proof together with the full statement. The missing line — the arc argument above — consumes only C_4 -freeness. A sweep of all 12 places where the upstream lemma is used confirmed that every one carries C_4 -freeness, so no downstream conclusion moved.

Corollary 7.20 (mass identity). *[proved here; machine-checked by w133_r8_q61.py; two-family review] In a C_4 -free P_6 -free graph with an induced C_6 , $t(v_i) = 0$ and*

$$a(v_i) = d(v_i) = 2 + [\text{the } (v_i, v_{i+3}) \text{ slot is occupied}].$$

Proof. By Lemma 7.18 the only off- Z neighbour a cycle vertex v_i can have is the vertex occupying the antipodal slot (v_i, v_{i+3}) , of which there is at most one by C_4 -freeness. $N(v_i) = \{v_{i-1}, v_{i+1}\}$ possibly with that vertex; $v_{i-1} \not\sim v_{i+1}$ (cyclic distance 2 on an induced hexagon), and the antipodal-slot vertex is adjacent to neither. So $N(v_i)$ is independent, $t(v_i) = 0$, and Lemma 2.1 gives $a = d$. $\square$

Over the exhaustive C_4 -free table on $6 \leq n \leq 7$ (111 508 connected graphs) plus 47 explicit and greedily generated carriers, restricted to the P_6 -free members carrying an induced C_6 , the classification of Lemma 7.18 and the identity of Corollary 7.20 were asserted on 1 330 hexagons, 1 300 off-hexagon vertices and 15 960 mass identities with 0 failures; ten hexagons carry two occupied antipodal slots simultaneously, so the occupied case is tested non-vacuously. [machine-verified: w133_r8_q61.py]

Lemma 7.21 (the frame collapses). *[proved here; machine-checked by w133_r8_q61.py; two-family review] Let $C \in \text{RES} \wedge (C6)$ with the hexagon $Z = (u, u_1, u_2, z, y, x)$ of Lemma 7.16. Then*

$$N(u) = \{u_1, x\}, \quad N(z) = \{u_2, y\}, \quad d(u) = d(z) = 2, \quad V(C) = Z \sqcup \{P?\} \sqcup \{Q?\} \sqcup W,$$

where P is the (at most one) vertex adjacent to u_1 and y , Q the (at most one) vertex adjacent to u_2 and x , and W the set of vertices with no Z -neighbour. Moreover every $w \in W$ is at distance 3 from both u and z , hence $\text{ecc}(w) = 3$, hence $a(w) = 2$ by (R3b) and Corollary 7.14: no vertex of W is a hub.

Proof. The antipodal pairs of Z are (u, z) , (u_1, y) and (u_2, x) . By Lemma 7.18, every off- Z vertex with a Z -neighbour realises one of these three pairs, and (u, z) is impossible since $\text{dist}(u, z) = 3$; each of the other two is realised by at most one vertex, since two would be two common neighbours of the pair. Call them P and Q . In particular no off- Z vertex is adjacent to u or to z , so $N(u) = N(u) \cap Z = \{u_1, x\}$ and $N(z) = \{u_2, y\}$. If w has no Z -neighbour then w is at distance ≥ 2 from u and shares no neighbour with u (both of $N(u)$'s members lie on Z), so $\text{dist}(u, w) = 3$ because $\text{diam}(C) = 3$; same at z . Thus $\text{ecc}(w) = 3$, and (R3b) gives $a(w) \leq 2$ while Corollary 7.14 applied to the distance-3 pair (u, w) gives $a(w) = 2$. $\square$

The four induced P_6 's that kill the four consecutive-pair slots explicitly are (u'_1, u, x, y, z, u_2) , (u'_2, z, y, x, u, u_1) , (A, u_1, u, x, y, z) and (B, x, u, u_1, u_2, z) , where u'_1, u'_2, A, B denote hypothetical occupants of the slots $(u, u_1), (u_2, z), (u_1, u_2), (y, x)$. Over 176 C_4 -free frames “hexagon + one consecutive-pair slot + an arbitrary edge pattern on the other five slots” the named witness is an induced P_6 every time. [machine-verified: w133_r8_q61.py]

Lemma 7.22. *[proved here; machine-checked by w133_r8_q61.py; two-family review] In $\text{RES} \wedge (C6)$, write $W_P := N(P) \cap W$. Then*

- (a) $N(w) \subseteq \{P, Q\} \cup W_P$ for every $w \in W_P$;
- (b) every $w \in W_P$ is adjacent to Q (so $W_P \neq \emptyset$ forces Q to exist);
- (c) $|W_P| \leq 1$, hence $d(P) \leq 4$ and $a(P) \leq 3$; symmetrically $a(Q) \leq 3$.

Proof. (a) (w, P, u_1, u, x) is an induced P_5 : w has no Z -neighbour; $P \not\sim u, x$ since P 's only Z -neighbours are u_1, y ; $u_1 \not\sim x$ on the induced hexagon. By (C6) no induced P_6 extends it, so every $v \in N(w) \setminus \{P\}$ is adjacent to one of P, u_1, u, x . Now $v \notin Z$ (as w has no Z -neighbour, v could only be a Z -vertex at distance 1 from w , which is excluded); $v \sim P$ puts $v \in W_P \cup \{Q\}$; $v \sim u_1$ puts $v \in \{P\}$ (the slot (u_1, u_2) is empty); $v \sim u$ is impossible since $N(u) \subseteq Z$; and $v \sim x$ puts $v \in \{Q\}$ (the slot (y, x) is empty).

(b) If $w \not\sim Q$ then by (a) $N(w) \subseteq \{P\} \cup W_P$, and $|N(w) \cap W_P| \leq 1$ since two W_P -neighbours of w would be two common neighbours of w and P . So either $N(w) = \{P\}$, or $N(w) = \{P, w'\}$ with $w' \in W_P \subseteq N(P)$ and hence $P \sim w'$; either way $N(w)$ is a clique, so $a(w) = 1$, contradicting $a(w) = 2$ from Lemma 7.21.

(c) Two vertices of W_P would be two common neighbours of P and Q by (b), a C_4 . Then $N(P) = \{u_1, y\} \sqcup (\{Q\} \cup W_P)$ with $u_1 \not\sim y$ (antipodal on an induced hexagon) and neither adjacent to anything else in $N(P)$, while $Q \sim w$; so $a(P) \leq 2 + 1 = 3$. $\square$

Theorem 7.23 (the hexagon branch is empty). *[proved here; machine-checked by `w133_r8_q61.py`, `w133_r9_frame_1.py`; **two-family review**] $\text{RES} \wedge (C6) = \emptyset$. Equivalently: every connected C_4 -free graph with $\text{rad} = 2$, $\text{diam} = 3$, $\ell > 3$ and (R3b) contains an induced P_6 .*

Proof. Suppose $C \in \text{RES} \wedge (C6)$ and set up the frame above. By Lemma 7.21 every hub of C lies in $\{u_1, u_2, x, y, P, Q\}$: the vertices u, z have $a = 2$, no vertex of W is a hub, and there is nothing else. By Corollary 7.20 and Lemma 7.22,

$$\begin{aligned} a(u_1) - 2 &= [P], & a(u_2) - 2 &= [Q], & a(x) - 2 &= [Q], & a(y) - 2 &= [P], \\ a(P) - 2 &\leq 1, & a(Q) - 2 &\leq 1, \end{aligned}$$

where $[P], [Q] \in \{0, 1\}$ record whether the slots are occupied. Hence

$$\sum_{h \in H} (a(h) - 2) \leq 3[P] + 3[Q] \leq 6.$$

But Lemma 7.7(a) demands $\sum_{h \in H} (a(h) - 2) > n$, and $n \geq |Z| = 6$, so the mass must be at least 7: a contradiction. $\square$

Remark 7.24 (the margin, and a second kill). *[proved here; machine-checked by `w133_r9_frame_1.py`; **two-family review**] The margin is not thin. By Lemma 7.8, $|H| \geq 6$, which forces all six candidate slots to be hubs, hence $[P] = [Q] = 1$, hence $n \geq 8$ and a required mass of ≥ 9 against a cap of exactly 6.*

Moreover the class dies a second time, by a route that never uses the mass inequality. By Lemma 7.21, $V = Z \sqcup \{P?\} \sqcup \{Q?\} \sqcup W$ and $N(u) = \{u_1, x\} \subseteq Z$. For $w \in W$, $\text{dist}(w, u) = 3$, so a shortest w - u path is w, a, b, u with $b \in \{u_1, x\}$ and $a \in N(u_1) \cup N(x) = \{u, u_2, P\} \cup \{u, y, Q\}$; adjacency to w forces $a \notin Z$, i.e. $a \in \{P, Q\}$. So $W = W_P \cup W_Q$, and Lemma 7.22(b) gives $W_P \subseteq W_Q$ and $W_Q \subseteq W_P$, whence $W = W_P$ and $|W| \leq 1$ and $n = 6 + [P] + [Q] + |W| \leq 9$. The enumeration of all frames $Z \sqcup \{P?\} \sqcup \{Q?\} \sqcup W$ with $|W| \leq 3$ and all edge patterns is therefore *exhaustive* for the class; it returns exactly 5 admissible frames, with

$$\begin{aligned} (P, Q, |W|=1) : n = 9, \ell = 8/3; & \quad (P, Q) : n = 8, \ell = 2.5; \\ (P) \text{ or } (Q) : n = 7, \ell = 16/7; & \quad (\text{neither}) : n = 6, \ell = 2. \end{aligned}$$

So $\max \ell = 8/3 < 3$, contradicting $\ell > 3$ directly. Both kills depend on §7.2; exactly one of them depends on the mass inequality.

Corollary 7.25. *[proved here; machine-checked by `w133_r8_q61.py`; **two-family review**] $C \in \text{RES}$ implies $\text{path}(C) \geq 6$.*

Proof. Theorems 7.13 and 7.23. $\square$

Proof of Theorem 7.1. Let G be a minimum-order counterexample to (6.1). By Proposition 6.4(i), $\text{rad}(G) \geq 2$; $\text{rad}(G) = 2$ is excluded by Corollary 5.5. If $\text{rad}(G) = 3$, Proposition 7.6 puts the core C in RES with (C6), contradicting Corollary 7.25 since $\text{path}(G) \geq \text{path}(C) \geq 6 = \text{rad}(G) + 3$. $\square$

Remark 7.26 (what is not reproduced). An earlier route to Theorem 7.23 ran through an explicit list of eleven possible slots for a hub and reduced the branch to a finite window $14 \leq n \leq 20$. Lemma 7.18 supersedes it: the window is closed by proof rather than by search, the intermediate lemmas become vacuous (their subjects u'_1, u'_2, A, B do not exist), and the window's lower endpoint was in any case withdrawn as unsupported by its own logs — the correct and sufficient bound is $n \geq |Z| = 6$. None of that material is reproduced here. Likewise a ball-confinement bound $|H| \leq 24$ and a centre identity, both unconditional and both used only by the superseded route, are omitted.

8 The band $[\ell] \geq 4$

The three results of this section carry the lower [machine-assisted proof, re-derived here; machine-checked by ; **one adversarial review pass**] tier of §1.3: the arguments originated in the free exploration channel, were re-derived line by line here (with the two repairs marked below), and passed *one* adversarial review pass, which returned no defect at any joint classified as mathematics. They are not on the same footing as §7.

The first two extend Lemma 4.7 downwards. Recall that its proof used $d \geq 5$ exactly twice, to keep the two extension vertices w and y far apart.

Lemma 8.1 ($d = 4$). [*machine-assisted proof, re-derived here; machine-checked by `w133_r10_adjudicate.py`, `w133_r11_adjudicate.py`; one adversarial review pass*] Let G be C_4 -free with a geodesic $u_0 u_1 u_2 u_3 u_4$, $a(u_0) \geq 3$, $a(u_4) \geq 2$, and a usable side-neighbour x of u_0 with $a(x) \geq 4$. Then $\text{path}(G) \geq 8 = d + 4$. No further hypothesis is needed.

Proof. Choose $y \in N(u_4)$ outside u_3 's component, so $y \not\sim u_3$.

Step 1: the kill accounting. $N(x)$ has at least $a(x) \geq 4$ components; u_0 's is one, and we let C_1, C_2, C_3 be three others with representatives w_1, w_2, w_3 . No neighbour of x is adjacent to u_4 , since $\text{dist}(w, u_4) \geq \text{dist}(u_0, u_4) - \text{dist}(u_0, w) = 4 - 2 = 2$ for $w \in N(x)$. Each of y, u_2, u_3 destroys at most one component, because x is adjacent to none of them — $x \not\sim u_1$ by usability; $x \not\sim u_2$, else u_0, u_2 have the common neighbours x, u_1 ; $x \not\sim u_3$ by distance; $\text{dist}(x, y) \geq \text{dist}(u_0, y) - 1 \geq 2$ using $\text{dist}(u_0, y) \geq \text{dist}(u_0, u_4) - 1 = 3$ — and C_4 -freeness gives $|N(x) \cap N(v)| \leq 1$ for every $v \not\sim x$.

Step 2: the standard path. If some w_i avoids all three kills, then $w_i, x, u_0, u_1, u_2, u_3, u_4, y$ is an induced P_8 . The chords: $w_i \not\sim u_0$ (different components of $N(x)$); $w_i \not\sim u_1$, else $w_i x u_0 u_1$ is a C_4 (using $x \not\sim u_1$); $w_i \not\sim u_2, u_3$ by assumption and $w_i \not\sim u_4$ by Step 1; $x \not\sim u_2, u_3, u_4, y$; the geodesic supplies $u_0 \not\sim u_2, u_3, u_4$, $u_1 \not\sim u_3, u_4$, $u_2 \not\sim u_4$; $y \not\sim u_3$ by choice; $y \not\sim u_2$, else u_2, u_4 have the common neighbours y, u_3 ; and $y \not\sim u_1, u_0$ by distance. All eight vertices are distinct: $w_i, x \in \text{ball}_2(u_0)$ with $w_i \neq u_2$ (as $x \not\sim u_2$), and $y \notin \{u_2, u_3\}$ since $y \not\sim u_3$ while $y \sim u_4$.

Step 3: all three kills land, on three distinct components. After relabelling, $w_y \sim y$, $w_2 \sim u_2$, $w_3 \sim u_3$ lie in three distinct components of $N(x)$, none of them u_0 's. Since $a(u_0) \geq 3$ we may pick $z \in N(u_0)$ outside both u_1 's and x 's components, so $z \not\sim u_1$ and $z \not\sim x$. Then $z \not\sim y$: otherwise $\text{dist}(u_0, y) \leq 1 + \text{dist}(z, y) = 2$, contradicting $\text{dist}(u_0, y) \geq 3$. Take

$$z - u_0 - x - w_y - y - u_4 - u_3 - u_2 \quad (8 \text{ vertices}).$$

Its edges are present by construction. The twenty-one non-adjacent pairs: $z \not\sim x$ (components of $N(u_0)$); $z \not\sim w_y$, else u_0, w_y have the common neighbours x, z ; $z \not\sim y$ (above); $z \not\sim u_4, u_3$ by distance; $z \not\sim u_2$, else u_0, u_2 have the common neighbours z, u_1 ; $u_0 \not\sim w_y$ (components of $N(x)$);

$u_0 \not\sim y$ by distance; $u_0 \not\sim u_4, u_3, u_2$ (geodesic); $x \not\sim y, x \not\sim u_4$ by distance, $x \not\sim u_3, u_2$ (Step 1); $w_y \not\sim u_4$ (Step 1); $w_y \not\sim u_3$ and $w_y \not\sim u_2$, because a vertex of $N(x)$ adjacent to u_3 (resp. u_2) must lie in the single component that u_3 (resp. u_2) kills, which is that of w_3 (resp. w_2), and w_y lies in a third; $y \not\sim u_3$ by choice; $y \not\sim u_2$ as in Step 2; $u_4 \not\sim u_2$ (geodesic). Distinctness: $z \neq x$ and $z \notin N(x)$; $w_y \in N(x)$ lies at distance 2 from u_0 and $w_y \neq u_2$ since $x \not\sim u_2$; $y \notin \{u_2, u_3\}$ as in Step 2. $\square$

Lemma 8.2 ($d = 3$, Case 1). *[machine-assisted proof, re-derived here; machine-checked by `w133_r10_adjudicate.py`, `w133_r11_adjudicate.py`; **one adversarial review pass**] Let G be C_4 -free with a geodesic $u_0 u_1 u_2 u_3$, $a(u_0) \geq 3$, $a(u_3) \geq 2$, a usable side-neighbour x of u_0 with $a(x) \geq 4$, and a usable side-neighbour y of u_3 with $y \not\sim x$. Then $\text{path}(G) \geq 7 = d + 4$.*

Proof. Fix such a y . $N(x)$ has ≥ 4 components, one of them u_0 's; pick representatives w_1, w_2, w_3 of three others. As in Lemma 8.1, each of y, u_2, u_3 kills at most one component, using $x \not\sim u_2$ (the C_4 through u_0, u_1), $x \not\sim u_3$ (distance) and $x \not\sim y$ (hypothesis).

Subcase 1a: some w_i avoids all three kills. Then $w_i, x, u_0, u_1, u_2, u_3, y$ is induced, by the chord list of Lemma 8.1 Step 2 read at $d = 3$: $w_i \not\sim u_0$ (component), $w_i \not\sim u_1$ (the C_4 through x, u_0), $w_i \not\sim u_2, u_3, y$ (kills avoided), $x \not\sim u_1, u_2, u_3, y$, $u_0 \not\sim u_2, u_3$, $u_1 \not\sim u_3$, $y \not\sim u_2$ (choice of y), $y \not\sim u_1$ (else u_1, u_3 have the common neighbours y, u_2), $y \not\sim u_0$ (distance).

Subcase 1b: the three kills land on three distinct components, say $w_1 \sim y$, $w_2 \sim u_2$, $w_3 \sim u_3$. Since $a(u_0) \geq 3$, pick $z \in N(u_0)$ outside both u_1 's and x 's components.

1b-i: $z \not\sim y$. Then $z, u_0, x, w_1, y, u_3, u_2$ is induced: $z \not\sim x$ (components of $N(u_0)$); $z \not\sim w_1$, else u_0, w_1 have the common neighbours x, z ; $z \not\sim y$ (assumed); $z \not\sim u_3$ (distance); $z \not\sim u_2$, else u_0, u_2 have the common neighbours z, u_1 ; $u_0 \not\sim w_1$ (components of $N(x)$); $u_0 \not\sim y$ (distance); $u_0 \not\sim u_3, u_2$ (geodesic); $x \not\sim y$ (hypothesis); $x \not\sim u_3$ (distance); $x \not\sim u_2$ (C_4); $w_1 \not\sim u_2, u_3$ (the u_2 - and u_3 -kills lie in other components); $y \not\sim u_2$ (choice of y).

1b-ii: $z \sim y$. Then $w_1, y, z, u_0, u_1, u_2, w_2$ is induced. Its edges are $w_1 y$ (the y -kill), yz (assumed), $zu_0, u_0 u_1, u_1 u_2, u_2 w_2$ (the u_2 -kill). The chords: $w_1 \not\sim z$, else u_0 and w_1 have the two common neighbours x, z ; $w_1 \not\sim u_0$ (components of $N(x)$); $w_1 \not\sim u_1$, else x, u_1 have the common neighbours w_1, u_0 ; $w_1 \not\sim u_2$ and $w_1 \not\sim w_2$ (different components of $N(x)$); $y \not\sim u_0$ (distance); $y \not\sim u_1$ (else u_1, u_3 have the common neighbours y, u_2); $y \not\sim u_2$ (choice of y); $y \not\sim w_2$ — *[first repaired step]* since $x \not\sim y$, C_4 -freeness gives $|N(x) \cap N(y)| \leq 1$ and w_1 already lies in it, so $w_2 \notin N(y)$; $z \not\sim u_1$ (components of $N(u_0)$); $z \not\sim u_2$ (else u_0, u_2 have the common neighbours z, u_1); $z \not\sim w_2$ (else u_0, w_2 have the common neighbours x, z); $u_0 \not\sim u_2$ (geodesic); $u_0 \not\sim w_2$, since $w_2 \in N(x)$ lies outside u_0 's component of $N(x)$; and $u_1 \not\sim w_2$ — *[second repaired step]*: the source text justified this by “ u_1 and w_2 share u_2 and are not adjacent”, which is not a reason; the correct reason is that $u_1 \sim w_2$ would give x and u_1 the two common neighbours u_0 and w_2 , a C_4 (note $x \neq u_1$ and $u_0 \neq w_2$). $\square$

The complementary case at $d = 3$ — every usable side-neighbour y of u_3 satisfies $y \sim x$ — is *not* covered by Lemma 8.2, and §9 shows that it is false as posed, with and without the natural repair hypothesis.

The third result is global.

Definition 8.3. Let G be connected and C_4 -free. A geodesic is *3-capped at the end u_0* if every usable side-neighbour x of u_0 has $a(x) \leq 3$ (vacuously so if there is none). G is *peripherally 3-capped* if every geodesic of length $\geq \text{rad}(G)$ is 3-capped at both of its ends.

Theorem 8.4. *[machine-assisted proof, re-derived here; machine-checked by `w133_r10_adjudicate.py`, `w133_r11_adjudicate.py`; **one adversarial review pass**] Every connected C_4 -free peripherally 3-capped graph satisfies $\ell(G) < 4$.*

Proof. Write $S := \{v : a(v) \geq 4\}$, $T := \{v : a(v) \leq 3\}$, $R := \text{rad}(G)$, and suppose for contradiction $\sum_v a(v) \geq 4n$.

Claim 1: every vertex v has at most 2 neighbours in S . Since $\text{ecc}(v) \geq R$ there is w with $\text{dist}(v, w) = \text{ecc}(v) \geq R$; any geodesic P from v to w has length $\geq R$ and is therefore 3-capped at v . The usable side-neighbours of v with respect to P are exactly $N(v) \setminus C_0$, where C_0 is the component of P 's first step in $N(v)$, and $|C_0| \leq 2$ by Lemma 2.1. An S -neighbour has $a \geq 4$ and so cannot be a usable side-neighbour; hence every S -neighbour of v lies in C_0 .

Claim 2: if v has two S -neighbours s_1, s_2 , then $s_1 \sim s_2$. By Claim 1's proof both lie in C_0 , and $|C_0| \leq 2$, so $C_0 = \{s_1, s_2\}$ is a matching edge of $N(v)$.

Claim 3: $G[S]$ is a disjoint union of triangles, edges and isolated vertices. By Claim 1, $\Delta(G[S]) \leq 2$, so the components are paths and cycles; Claim 2 forces the two S -neighbours of any degree-2 vertex of $G[S]$ to be adjacent, and the resulting chord raises a degree to 3 unless the component is a triangle or smaller.

Let k_3, k_2, k_1 count the K_3 -, K_2 - and K_1 -components of $G[S]$, so $|S| = 3k_3 + 2k_2 + k_1$.

Claim 4. For a K_3 component $\{a, b, c\}$, no T -vertex is adjacent to two of a, b, c : if u is adjacent to a and b , then a, b are two common neighbours of u and c . *Claim 5.* For a K_2 component $\{a, b\}$, at most one T -vertex is adjacent to both, for the same reason. *Claim 6.* If $u \in T$ has two S -neighbours they are the endpoints of a K_2 component: by Claim 2 applied to u they are adjacent, hence form an edge of $G[S]$, which lies in a K_2 or a K_3 component, and K_3 is excluded by Claim 4.

Upper bound. Partition T by $d_S(u) \in \{0, 1, 2\}$ into T_0, T_1, T_2 . By Claims 5 and 6, $|T_2| \leq k_2$, so $E(S, T) = 2|T_2| + |T_1| \leq k_2 + |T| = k_2 + n - |S|$.

Lower bound. For $v \in S$ split $t(v)$ into $t_{SS} + t_{ST} + t_{TT}$ by the location of the pair. If v lies in a K_3 then $d_S(v) = 2$ and those two neighbours are matched to each other in $N(v)$, so $t_{SS}(v) = 1$, $t_{ST}(v) = 0$ and $a(v) = (2 + d_T(v)) - (1 + t_{TT}(v))$, giving $d_T(v) \geq a(v) - 1$. If v lies in a K_2 then $d_S(v) = 1$, $t_{SS}(v) = 0$ and $t_{ST}(v) \leq 1$, giving again $d_T(v) \geq a(v) - 1$. If v is a K_1 then $d_S(v) = 0$ and $d_T(v) \geq a(v)$. Summing, with $|S| - k_1$ the number of S -vertices in K_3 - and K_2 -components, $E(S, T) = \sum_{v \in S} d_T(v) \geq \sum_{v \in S} a(v) - (|S| - k_1)$. Bounding each T -vertex's a -value by 3 in $\sum_v a(v) \geq 4n$ gives $\sum_{v \in S} a(v) \geq 4n - 3(n - |S|) = n + 3|S|$, hence $E(S, T) \geq n + 2|S| + k_1$.

The contradiction. Combining, $n + 2|S| + k_1 \leq k_2 + n - |S|$, i.e. $3|S| + k_1 \leq k_2$, i.e. $9k_3 + 5k_2 + 4k_1 \leq 0$, so $k_3 = k_2 = k_1 = 0$ and $S = \emptyset$. But then every $a(v) \leq 3$ and $\sum_v a(v) \leq 3n < 4n$. $\square$

Corollary 8.5. *[machine-assisted proof, re-derived here; machine-checked by `w133_r10_adjudicate.py`; **one adversarial review pass**] Every connected C_4 -free graph with $\ell(G) > 4$ has, at every radius, a geodesic of length $\geq \text{rad}(G)$ with an end u_0 carrying a usable side-neighbour x with $a(x) \geq 4$.*

The lemma chain of Theorem 8.4 was machine-tested non-vacuously: 449 peripherally 3-capped graphs in the sweep, 60 of them with $S \neq \emptyset$, with K_1 -, K_2 -, K_3 -components and a T_2 straddler all realised, all passing every claim and both bounds. Consistency was checked on the two natural near-misses: the incidence graph of $PG(2, 3)$ ($n = 26$, 4-regular, $\ell = 4.0$) and that of $PG(3, 2)$ ($n = 50$, $\ell = 4.2$) are both verified *not* peripherally 3-capped. [machine-verified: `w133_r10_adjudicate.py`]

Remark 8.6 (the boundary, and a vacuity annotation). Run with strict inequalities the chain already yields $9k_3 + 5k_2 + 4k_1 < 0$ without the $S = \emptyset$ endgame; the non-strict run plus that endgame is exactly what buys the boundary case $\ell = 4$, so the conclusion is $\ell < 4$ and not merely $\ell \leq 4$. The review pass confirmed the boundary run independently. At the time Theorem 8.4 was recorded, no connected C_4 -free graph with $\ell > 4$ and $\text{rad} \geq 5$ was known to the campaign, so the $\text{rad} \geq 5$ slice of Corollary 8.5 carried an explicit *possibly vacuous* annotation; the theorem itself is not vacuous, since it executes on the $\text{rad} = 3$ class. That annotation was later removed by an explicit construction which has not been through the review protocol and is therefore not reported here.

9 Negative results, refutations, and named obstacles

This section is part of the report, not an appendix to it: each item closes a route, and three of them close routes that the campaign itself had been pursuing.

9.1 No two-endpoint lemma reaches $\text{rad} + k$ for $k \geq 4$

Proposition 9.1. *[machine-verified: `wowii133_gplus.c`] For $m \geq 1$ let T_m be the tree consisting of a path $u - s - v$ with m pendant vertices attached to u and m to v . Then T_m is C_4 -free with $n = 2m + 3$, $\text{rad}(T_m) = 2$, $a(u) = a(v) = m + 1$, $\ell(T_m) = (4m + 4)/(2m + 3) \rightarrow 2$, and $\text{path}(T_m) = 5$ for every m . Consequently, for every $k \geq 4$ the statement*

“ C_4 -free with a pair at distance rad whose a -values are both $\geq k$ implies $\text{path} \geq \text{rad} + k$ ”

is false, and no strengthening of Lemma 4.2 that reads only the two endpoints of a geodesic can prove (C133-b) in the band $\lfloor \ell \rfloor \geq 4$.

Proof. T_m is a tree, hence C_4 -free. $\text{ecc}(s) = 2$ and $\text{ecc}(u) = 3$, so $\text{rad} = 2$ and $\text{dist}(u, v) = 2 = \text{rad}$. The a -values are $m + 1$ at u and at v , 2 at s and 1 at each of the $2m$ pendant vertices, giving $\sum a = 4m + 4$. A longest induced path is $\text{leaf} - u - s - v - \text{leaf}$, on 5 vertices. For $k \geq 4$ and $m \geq k - 1$ the hypothesis holds while $\text{rad} + k = 2 + k \geq 6 > 5 = \text{path}$. $\square$

Note that T_m does *not* contradict the conjecture: $\lfloor \ell \rfloor = 2$ and $\text{path} = 5 \geq \text{rad} + 2$. What it kills is a proof technique. It also shows why Theorem 4.5 is stated with the hypothesis $\mu \geq 2$ rather than with a hypothesis on the endpoints only: $\mu(T_m) = 1$.

9.2 The layered-counting route dies above radius 2

Theorem 5.3 proves the statement (GP_2) : C_4 -free with $\text{rad} = 2$ and $\ell \geq 3$ implies there is a pair at distance $\geq \text{rad}$ with a -values $(\geq 3, \geq 2)$. The obvious next step is (GP_3) .

Proposition 9.2. *[machine-verified: `w133_r3_counterexample.py`] (GP_3) is false. Let A be the Hoffman–Singleton graph with two pendant vertices attached to every vertex. Then A is connected and C_4 -free with $n = 150$, $m = 275$, $\text{rad} = 3$, $\text{diam} = 4$, $\sum_v a(v) = 550$, $\ell = 11/3$ and $\lfloor \ell \rfloor = 3$ — so A lies in the band under attack — while the number of pairs at distance $\geq \text{rad}$ with a -values $(\geq 3, \geq 2)$ is zero: the largest distance between two vertices of a -value ≥ 2 is $2 < 3 = \text{rad}$. The graph B obtained by attaching only three pendant vertices, at three mutually distance-2 vertices with no common neighbour, has $n = 53$, $\text{rad} = 3$, $\lfloor \ell \rfloor = 6$ and again zero such*

pairs. Both satisfy the conjecture (A has an induced path on ≥ 16 vertices against a requirement of 6).

Verification. The vertices of a -value ≥ 2 are exactly the 50 Hoffman–Singleton vertices, which lie pairwise within distance $\text{diam}(\text{HS}) = 2$; every vertex at distance ≥ 3 from anything is a pendant vertex, of a -value 1. The invariants are computed by the named script. $\square$

The consequence is sharper than it looks. The route that proves Theorem 5.3 aims to show “ $(\star)$ and $\text{rad} \geq 3$ and a high vertex exists $\Rightarrow 3n - \sum a > 0$ ”; the witness A satisfies $(\star)$, has $\text{rad} = 3$ and 50 high vertices, and has $3n - \sum a = 450 - 550 = -100 < 0$. So the fibre count of Theorem 5.3 cannot be generalised to radius 3, and any proof there must genuinely construct an induced path rather than count. Only three pendant vertices suffice to lift the radius from 2 to 3 and destroy every good pair.

9.3 The deep endpoint lemma cannot be pushed to $d = 3$

Lemma 8.1 extends Lemma 4.7 to $d = 4$ with no extra hypothesis, and Lemma 8.2 extends it to $d = 3$ under the hypothesis that some usable side-neighbour y of u_3 satisfies $y \not\sim x$. The complementary configuration — call a frame *forced* if every usable side-neighbour y of u_3 satisfies $y \sim x$ — is not merely unproved. It is false, twice over.

Proposition 9.3 (the bare statement is false). *[machine-verified: w133_r10_adjudicate.py] Let G_1 be the graph on $\{0, \dots, 9\}$ with edges*

$$09, 05, 15, 23, 25, 26, 28, 49, 46, 57, 58.$$

Then G_1 is connected and C_4 -free; $2 - 6 - 4 - 9$ is a geodesic of length 3; $a(2) = 3$ and $a(9) = 2$; $x = 5$ is a usable side-neighbour of $u_0 = 2$ with $a(5) = 4$; the frame is forced (the unique usable side-neighbour of $u_3 = 9$ is 0, and $0 \sim 5$); and $\text{path}(G_1) = 6 < 7 = d + 4$. Hence the $d = 3$ analogue of Lemma 8.1 is false as stated.

The vertices of a -value 1 in G_1 are the leaves 1, 3, 7 and the triangle-leaf 8; peeling them leaves a 6-cycle, in which the frame no longer exists. This is what made the following “ $F4$ form” — the same statement with the extra hypothesis $\mu(G) \geq 2$, which is free in the intended application because peeling preserves C_4 -freeness and raises the slack (Lemma 6.2) — look like the right repair. It is not.

Theorem 9.4 (the repaired statement is also false). *[machine-verified: w133_r11_adjudicate.py] Let H be the graph on $\{0, \dots, 9\}$ with edges*

$$01, 04, 05, 08, 12, 23, 26, 35, 37, 39, 46, 47, 48, 89.$$

Then H is connected and C_4 -free with a -value vector $(3, 2, 3, 4, 3, 2, 2, 2, 2, 2)$, so $\mu(H) = 2$ and no peeling is possible; $2 - 1 - 0 - 8$ is a geodesic of length 3 with $a(u_0) = 3$ and $a(u_3) = 2$; $x = 3$ is a usable side-neighbour of u_0 with $a(x) = 4$; the frame is forced, the unique usable side-neighbour of $u_3 = 8$ being 9, with $9 \sim 3$; and $\text{path}(H) = 6 < 7$, witnessed by the induced path $0 - 8 - 9 - 3 - 2 - 6$ and by exhaustive search for the upper bound.

Verification, from the edge list alone. $N(0) = \{1, 4, 5, 8\}$ with the single internal edge 48, so $a(0) = 3$; $N(2) = \{1, 3, 6\}$ is independent, so $a(2) = 3$ and every neighbour of 2 outside $u_1 = 1$ ’s

component is usable; $N(3) = \{2, 5, 7, 9\}$ is independent, so $a(3) = 4$; $N(8) = \{0, 4, 9\}$ with the single internal edge 04, so $a(8) = 2$ and the usable side-neighbour set of 8 with respect to the geodesic is exactly $\{9\}$, and 39 is an edge. No two vertices have two common neighbours. Neither 2 nor 8 has a neighbour in common with the other, and $2 - 1 - 0 - 8$ is a shortest path. An exhaustive enumeration of induced paths returns 6 as the maximum. $\square$

Remark 9.5 (scope, and why G_1 and H do not touch §8). $\sum_v a(v) = 25$ and $n = 10$ give $\ell(H) = 2.5 < 3$, so H lies *outside* the live class of §6, which carries $\ell > 3$. What H refutes is therefore the hypothesis-free *local* route at $d = 3$ — the deep endpoint extension cannot be pushed down to $d = 3$ in this currency, with or without $\mu \geq 2$ — and not Problem 6.6. The next attempt at $D = 3$ must consume either a global hypothesis such as $\ell > 3$ or a declared a -value surcharge. Neither Lemma 8.1 nor Lemma 8.2 is touched: $\text{diam}(H) = 3$, so H carries no geodesic of length ≥ 4 and Lemma 8.1 cannot execute; and an exhaustive enumeration of H 's qualifying $d = 3$ frames returns exactly two, $2 - 1 - 0 - 8$ and $2 - 6 - 4 - 8$, both forced, so Lemma 8.2 cannot execute either.

Remark 9.6 (a methodological correction). Theorem 9.4 was found by the adversarial reviewer assigned to §8, on a target that the campaign's own sweep had recorded as open after finding no counterexample among 1112 *forced frames* (of which 63 satisfied $\mu \geq 2$). The sweep was frame-local: it enumerated local skeletons and could not see $\text{path}(G)$ of a completed graph. The lesson, recorded here because the same species of error appears twice in the campaign's logs, is that “0 counterexamples in N frames” is evidence about frames and never about graphs, and must not be written in a way that reads otherwise.

9.4 Named obstacles

Proposition 9.7 (surgery). *[machine-verified: wowii133_families.py] Let G^* be the Hoffman–Singleton graph with one edge subdivided. Then G^* is C_4 -free with $n = 51$, $\delta = 2$, $\text{rad} = 3$, $\ell = 6.902$ and $\lfloor \ell \rfloor = 6$, so the conjecture demands $\text{path} \geq 9$ (true: $\text{path} \geq 20$) while $\max(\text{rad} + 2, \delta + 2, \text{rad} + \lfloor \delta/2 \rfloor) = 5$. Hence Theorems 3.4, 3.6 and 3.7 cannot be combined into a proof of the conjecture.*

Every bound in §3 is driven by δ or rad ; both are destroyed by one local modification while ℓ moves by $O(1/n)$. A correct proof must use a quantity as robust as ℓ .

Three further obstacles are recorded without proof, since each is an observation rather than a theorem. (O2) *The degeneracy factor 2*. Every greedy argument really bounds path by $2 + \max_{H \subseteq G} \delta(H)$, i.e. by the degeneracy, which can be as small as half the average degree, whereas the conjecture needs about the average degree; closing that factor for C_4 -free graphs is a sharp form of the known results on induced subdivisions in $K_{s,s}$ -free graphs of large average degree, for which only tower-type bounds are available. (This attribution is from memory: the campaign ran without web access and it was not checked against the literature.) (O3) *No per-vertex charging*. The star $K_{1,m}$ has $a(\text{centre}) = m$ but $\text{path} = 3$, so no bound of the form “ $\text{path} \geq f(a(v)) + g(v)$ for some vertex v ” can exist: any proof must genuinely average. (O4) *Sharpness at the bottom*. By Proposition 3.9 five graphs attain equality, so no argument may lose even one vertex.

Finally, one negative result that saves computation:

Proposition 9.8. *[proved here] Every circulant graph of degree ≥ 4 contains a 4-cycle, since a connection set containing $\pm a$ and $\pm b$ with $a \neq b$ gives $0 = a + (-a) = b + (-b)$ and hence two*

vertices with two common neighbours. Circulants are therefore useless as test samples for any C_4 -free statement with a degree floor ≥ 4 .

10 Status and open problems

The conjecture is open. What remains, in the order in which the campaign ranks it:

1. **Problem 6.6 at $D = 3$.** By Theorem 7.1, a minimum-order counterexample in the band $[\ell] = 3$ has radius ≥ 4 , and §6 reduces radius ≥ 4 to Problem 6.6 at $D = \text{rad} - 1 \geq 3$. The case $D = 2$ is Theorem 6.7; $D = 4$ and $D \geq 5$ are available from Lemma 8.1 and Lemma 4.7 *under the side-neighbour hypothesis* $a(x) \geq 4$, which still has to be discharged from $\ell > 3$. The case $D = 3$ is the first genuinely open one, and Theorem 9.4 says the local route to it is closed: a global hypothesis must be paid for.
2. **Discharging the side-neighbour hypothesis.** Corollary 8.5 discharges exactly this hypothesis, but from $\ell > 4$ rather than $\ell > 3$. An analogue at $\ell > 3$ would combine with Lemmas 4.7 and 8.1 to give Problem 6.6 for $D = 4$ and $D \geq 5$ outright.
3. **The band $[\ell] \geq 4$.** Corollary 8.5 supplies a peripheral geodesic with a high side-neighbour at every radius; what a proof still needs at that geodesic is $a(u_0) \geq 3$ and $a(u_d) \geq 2$. Further reductions of the $\text{rad} \geq 5$ slice were obtained after the results reported here but had not completed the review protocol of §1.3 at the time of writing, and are deliberately not stated.
4. **A7.** Proposition 3.10 records the real-valued strengthening $\ell(G) \leq \text{path}(G) - \text{rad}(G)$, verified over the whole exhaustive range and tight on the same five graphs. Proving it would prove the conjecture, and Theorem 6.7 is a proved non-trivial instance of it. It remains the campaign’s recommended target.
5. **Formalisation.** Nothing here is formalised. The one convention that a Lean formalisation must not lose is the strong reading of C_4 -freeness (§1): several lemmas above — Lemma 7.10(a) among them — are false under the induced-only reading.

A Verification artifacts

Every script named below lives in the campaign repository at the stated relative path; each asserts the statements attributed to it and exits non-zero on the first violation. Logs are the same filename with extension `.out`. No result in this paper was obtained with a SAT solver.

Artifact	What it certifies
notes/proofs/ wowii133_-verify.c	The exhaustive generator and assertion harness: all connected graphs on $n \leq 10$ (136 362 861 evaluations) and all connected C_4 -free graphs on $n \leq 13$ (13 544 784); asserts Theorems 3.3, 3.4, 3.6, 3.7 and the coverage census of §3.1; self-validates its class counts against the number of connected graphs on n nodes.

Artifact	What it certifies
notes/proofs/families.py	wowii133_- The 79 structured graphs up to $n = 183$ (Petersen, Hoffman–Singleton, incidence graphs of $PG(2, q)$, polarity graphs ER_q , cages, random girth-5 regular graphs), their invariants, and the surgery witness of Proposition 9.7.
notes/proofs/sa.c	wowii133_- The adversarial simulated-annealing search over connected C_4 -free graphs, $n = 10..24$, minimising path – rad – $\lfloor \ell \rfloor$ ($\approx 2 \cdot 10^6$ proposals, minimum slack 1 for $n \geq 14$).
problems/wowii/wowii133_gplus.c	The same harness extended with assertions for Lemmas 4.1, 4.2, 4.3 (including the 334 failures of the $d = 3$ variant), 2.2, Theorem 4.5, Lemma 7.16, the structural steps of Theorem 6.7, and the hexagon-rigidity assertions (1272 non-vacuous induced- C_6 instances); the coverage census $606 \rightarrow 0$.
problems/wowii/w133_r3_counterexample.py	Proposition 9.2 (witnesses A and B) and the residual analysis of §6; the frames backing Lemma 4.7.
problems/wowii/w133_g12_nonempty.py	Proposition 6.9, triangle-free branch: Hoffman–Singleton lies in Theorem 6.7’s hypothesis class, with an induced-path certificate.
problems/wowii/w133_res3_search.py	Proposition 6.9, triangle branch (ER_5, ER_7, ER_{11}); the nine witnesses of Remark 7.3 and the verification that all nine violate (R3b).
problems/wowii/w133_res_kill.py	Remark 7.4 (the five members of $RES(a) \setminus RES$); Lemmas 7.7, 7.8, 7.9, 5.1, 5.2, Theorem 5.3 over a 48-graph pool (1363 distance-3 pairs, 329 subsets, 371 eccentricity-2 vertices, 301 centres).
problems/wowii/w133_r6_adjudicate.py	Lemma 7.11 for $k = 3, 4, 5$ (4226/67 344/194 280 endpoints) and the friendship-graph tightness; the star counterexample that establishes the gap repaired by Step 1 of Theorem 6.7; the classification of triangle-free diameter-2 C_4 -free graphs on $n \leq 7$ as 25 stars plus 12 Moore graphs.
problems/wowii/w133_r7_friend.py	Lemma 7.12 and the layer decomposition of Lemma 7.10 on 79 852 triangular-endpoint carriers, plus the two bounded integer sweeps used in Theorem 7.13.
problems/wowii/w133_r8_q61.py	Lemmas 7.17, 7.18, Corollary 7.20 (1330 hexagons, 1300 classified off-hexagon vertices, 15 960 mass identities); Lemma 7.21’s 176 frames; the exhaustive frame enumeration behind Theorem 7.23, whose realised maximum mass is exactly 6.
problems/wowii/w133_r9_frame_l.py	The five admissible frames of Remark 7.24 with ℓ computed; $\max \ell = 8/3$.
problems/wowii/w133_r9_controls.py	The verified control graphs: Petersen, the $PG(3, 2)$ incidence graph ($n = 50$, $\ell = 4.2$, $\text{rad} = 3$) and the $PG(2, 3)$ incidence graph ($n = 26$, $\ell = 4.0$), with exact induced-path search.

Artifact		What it certifies
problems/wowii/ r10_adjudicate.py	w133_-	Proposition 9.3; the $d = 3$ and $d = 4$ frame sweeps; the lemma chain of Theorem 8.4 on 449 peripherally 3-capped graphs; the verification that $PG(2, 3)$ and $PG(3, 2)$ are not peripherally 3-capped.
problems/wowii/ r11_adjudicate.py	w133_-	Theorem 9.4 re-verified from the edge list by independent code, including $\text{path}(H) = 6$ by exhaustion, $\ell(H) = 2.5$, and the enumeration of H 's two qualifying $d = 3$ frames; Remark 6.3's peeling arithmetic on G_1 .
problems/wowii/ judge_*.py	w133_r8_-	The first reviewer's own verification code for §7.2, written independently of the campaign's: an exhaustive pool of all 4448 connected C_4 -free graphs on $n \leq 10$ whose class counts independently reproduce the campaign's, plus explicit families; $\approx 90\,000$ assertions, 0 failures.
problems/wowii/ judge20_*.py	w133_r8_-	The blind reviewer's own verification code for §7.3, including the counterexample of Remark 7.19 and a re-verification of the repaired Lemma 7.18 over 2 105 408 frames.
problems/wowii/ R23_opus.md	w133_S3_-	The adversarial review report on §8, containing the refutation H of Theorem 9.4 and the peeling-arithmetic defect of Remark 6.3.
prompts/ w133_S3_R23.md		The self-contained brief placed under that review: the statements and full proofs of Lemmas 8.1, 8.2 and Theorem 8.4 exactly as reviewed, with the named joints and the mandatory probes.
problems/wowii/ GDIFF_qwen.md	w133_S3_-	The diff-scoped second review pass over §7.3 and the repair of Remark 7.19, returning no mathematics defect.
problems/wowii/ G12_qwen.md	w133_S3_-	The review pass that located the gap in Theorem 6.7 Step 1, repaired there.
notes/proofs/ draft.md	wowii133_-	The campaign's running attack report: the full round-by-round record, including every adjudication, every retraction, and the material omitted from this paper.